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Overview of Neuroembryology

Part 1: Embryonic development and spinal cord

Karen Poole, PhD
Department of Biomedical and Anatomical Sciences
kpoole@nyit.edu

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Make.
Heal.
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Session Objectives

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1. Explain the sequence of events and the timing of neural tube formation in a human embryo.
2. Describe the cellular organization of the neural tube as well as neural crest derivatives.
3. Compare and contrast the development of the spinal cord with that of the brain.
4. Correlate the major brain developmental landmarks (e.g. vesicles) with the anatomy of the human brain at birth.
5. Apply basic neurodevelopmental considerations to the analysis of the most common neuroanatomical malformations at birth.

Source: Course Syllabus

Quick Review of Gastrulation

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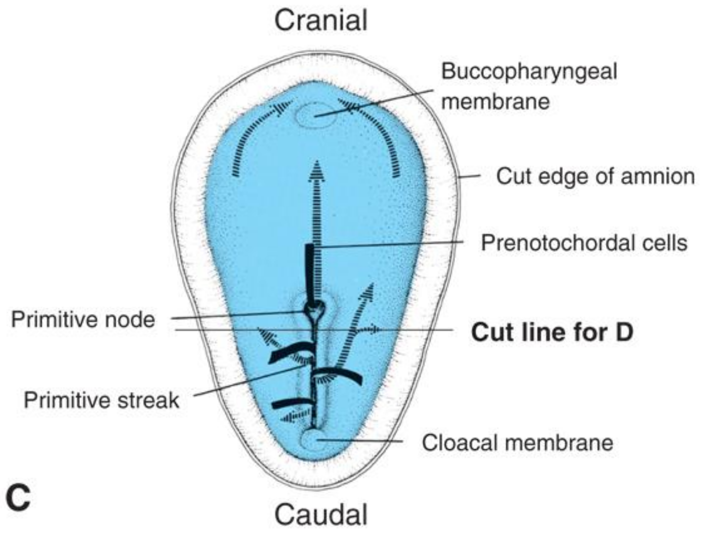
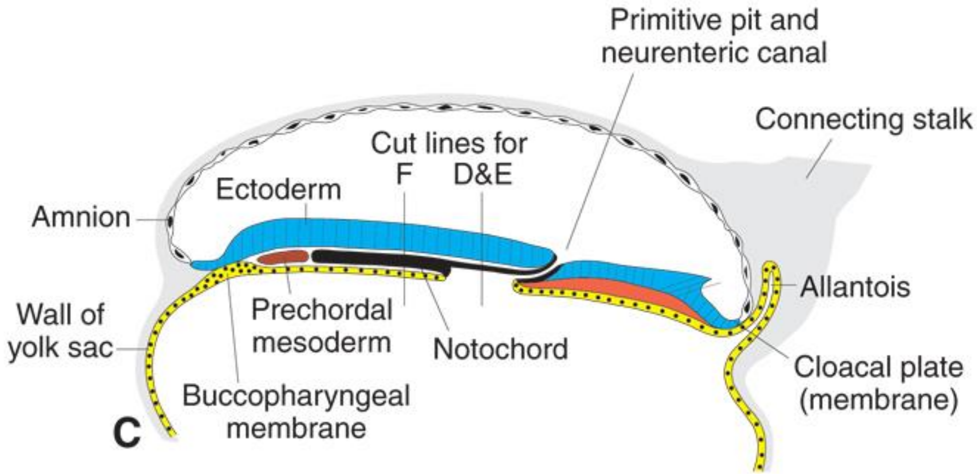
Note: times listed in this lecture are from fertilization (which is offset about 2 weeks from clinical definition of pregnancy)



2-4C Gastrulation: Primitive node formation



2-5C Gastrulation: Fate map and notochord

Neural Plate: The Beginning of the Nervous System

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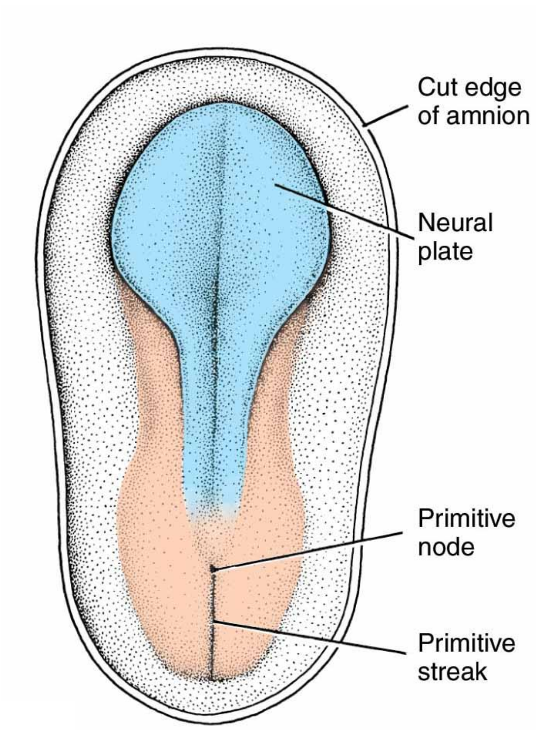


19 Days

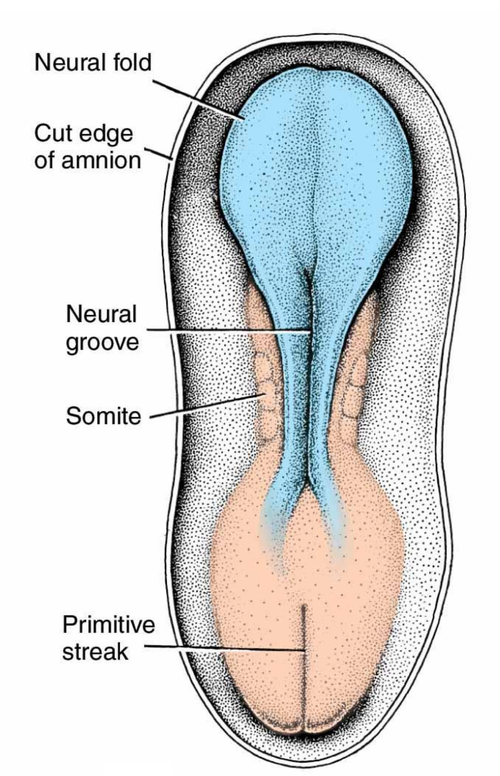
The notochord and paraxial mesenchyme induces the overlying ectoderm to thicken into the neural plate.

This begins rostrally and progresses caudally.

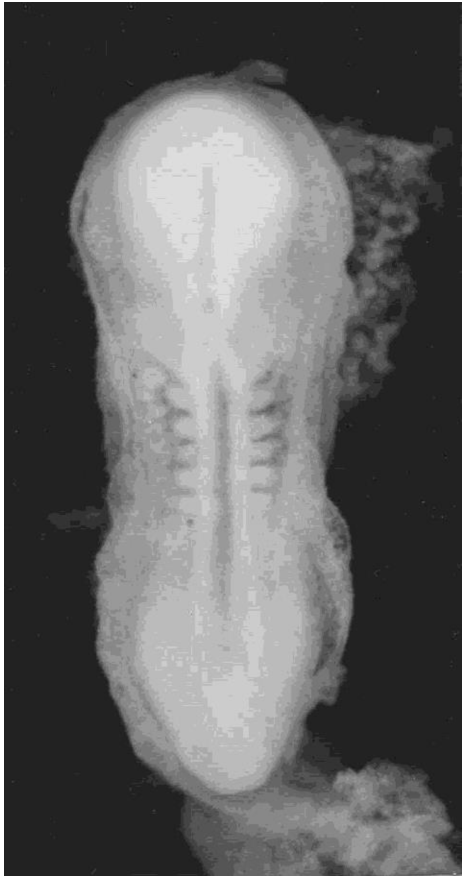
Involves many signaling molecules, including:

- transforming growth factor-β
- Wnts
- sonic hedgehog (SHH)
- bone morphogenic proteins (BMPs)

NEURULATION



20 Days



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The midline of the neural plate folds inward, creating a neural groove, with a neural fold on either side.

Note the primitive streak present here—neurulation and gastrulation overlap (weeks 3-4 of development).

NEURULATION

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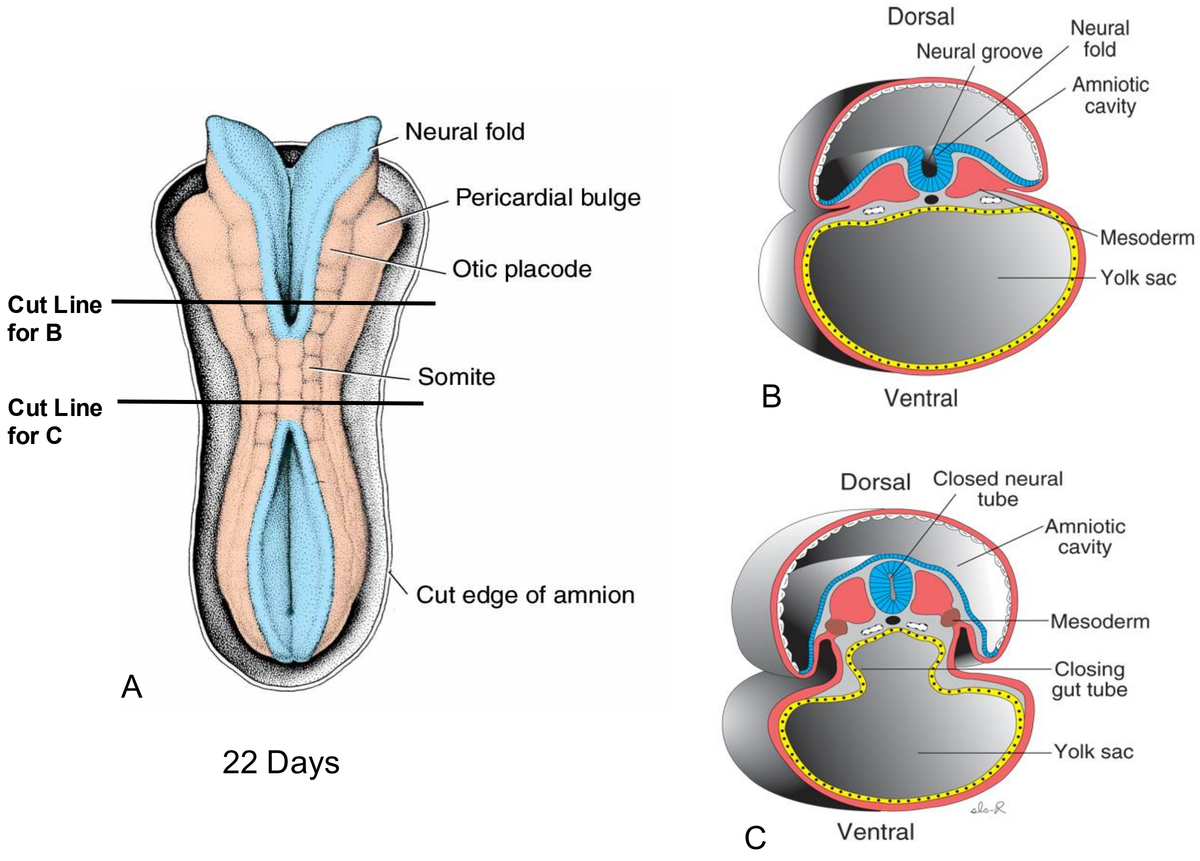

3-2C Neurulation: Neural tube closure

NEURULATION (CROSS-SECTIONAL VIEW)

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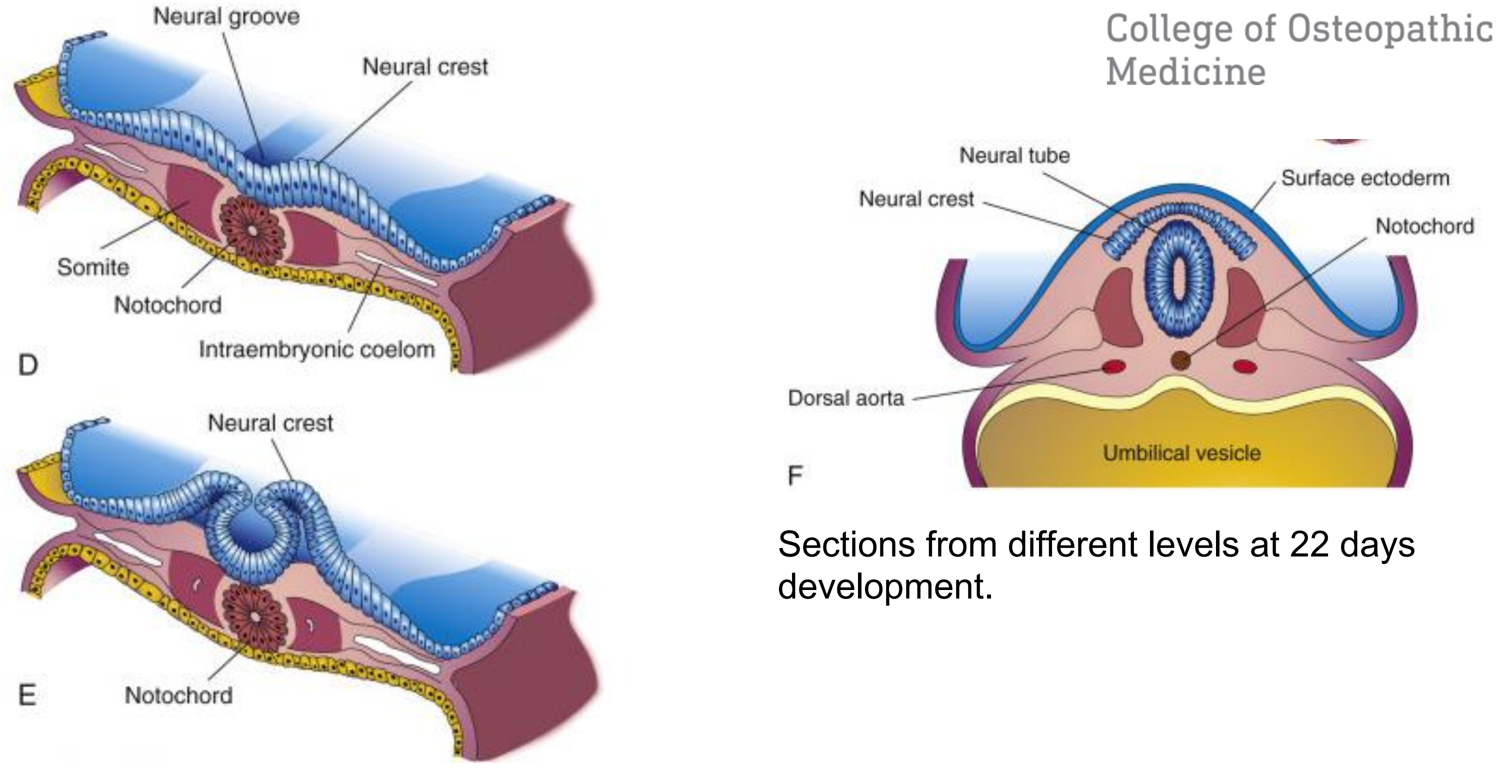

Sections from different levels at 22 days development.

Moore, The Developing Human, Fig. 17.1

NEURULATION

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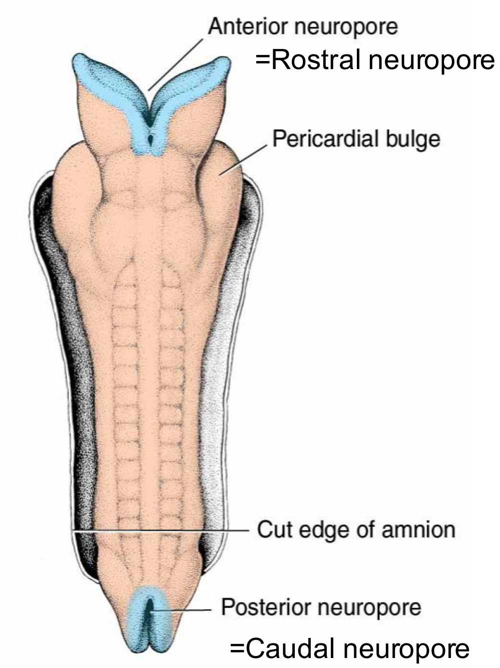
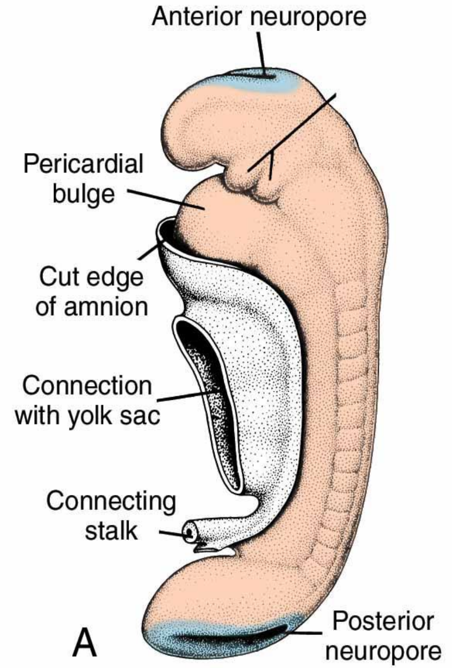
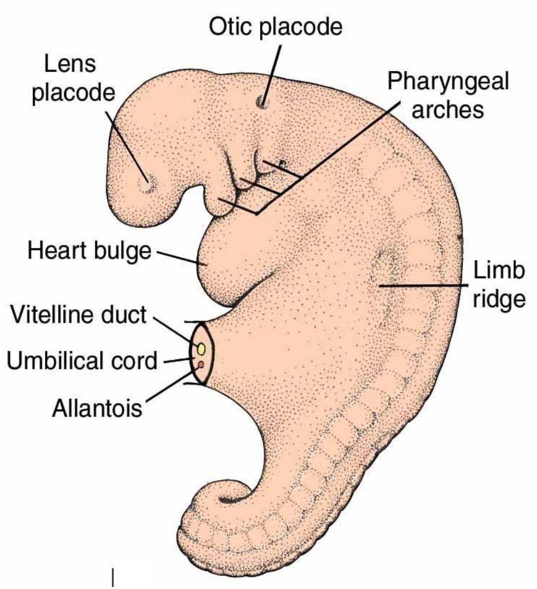


23 Days



25 Days

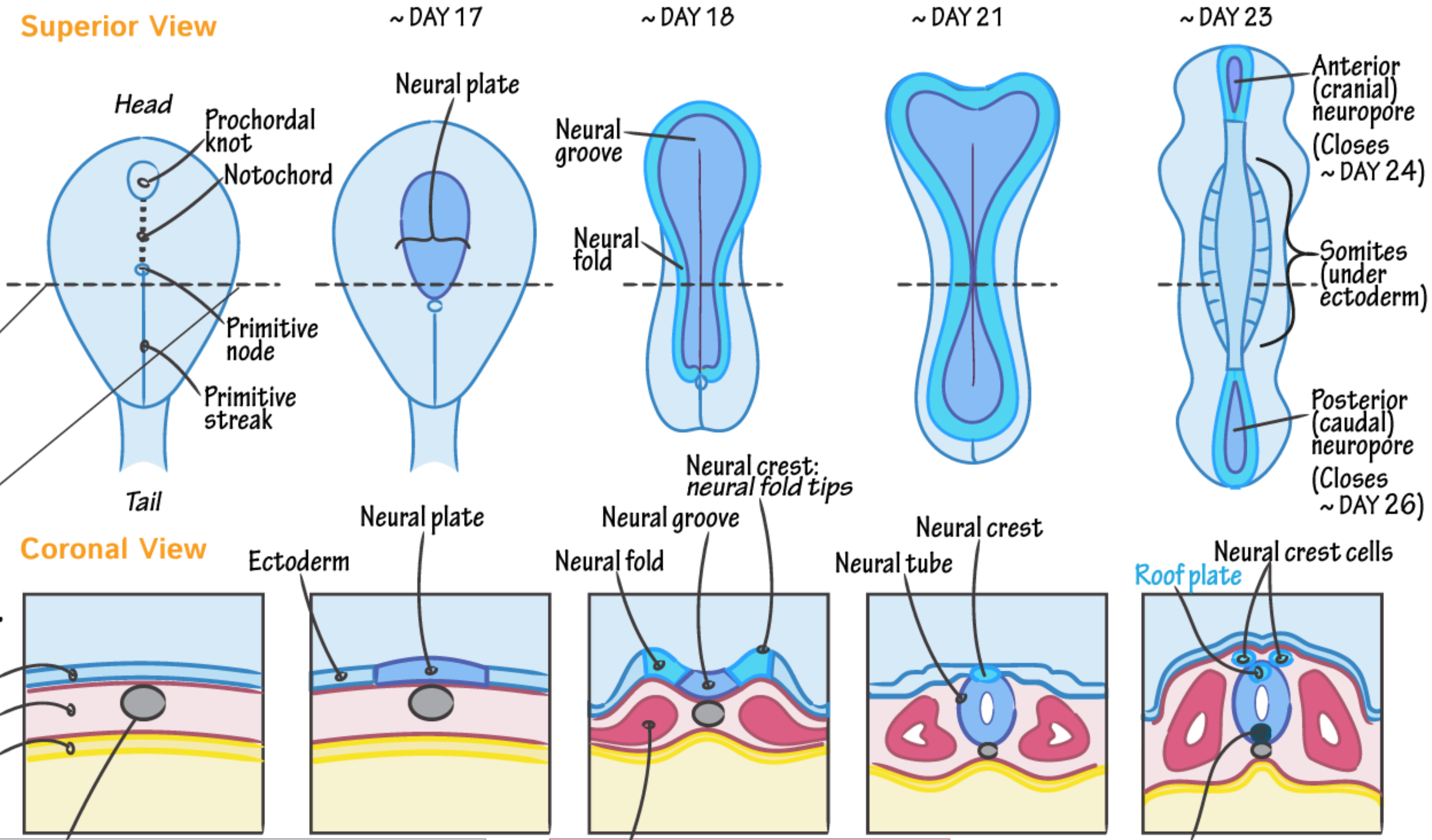


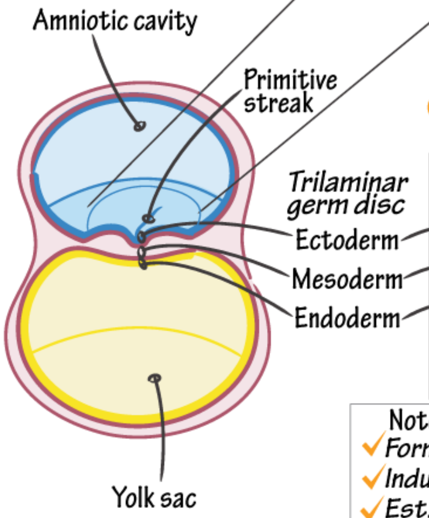
28 Days

NEURULATION

Neurulation

- ✓ Notochord induces overlying ectoderm to develop into the neural plate.
- ✓ The neural plate folds into the neural tube and the neural crests are pinched off.
- ✓ The neural tube derives the CNS.
- ✓ The neural crests derive the PNS + select other cells (eg, melanocytes).

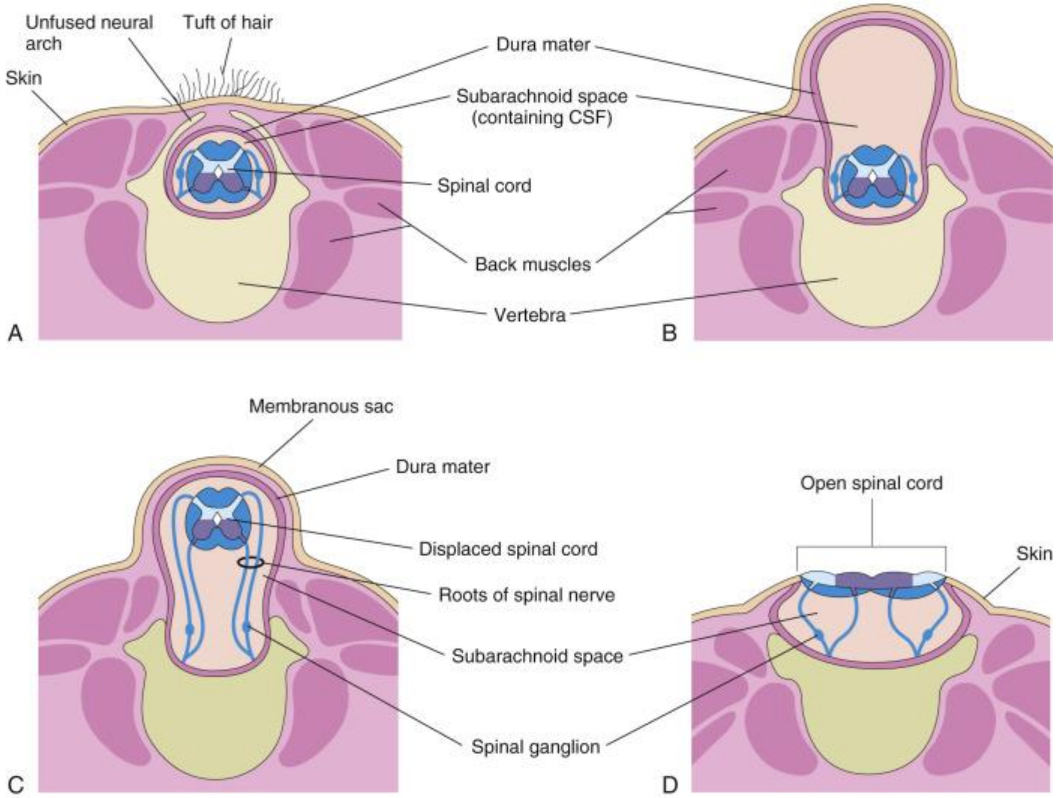




Notochord
- ✓ *Forms the embryonic central axis.*
- ✓ *Induces neural plate formation.*
- ✓ *Est. spine: degenerates to nucleus pulposus.*

Somite (*Paraxial mesoderm*)
- ✓ *Sclerotome (bone and cartilage)*
- ✓ *Myotome (skeletal muscle)*
- ✓ *Dermatome (dermis)*

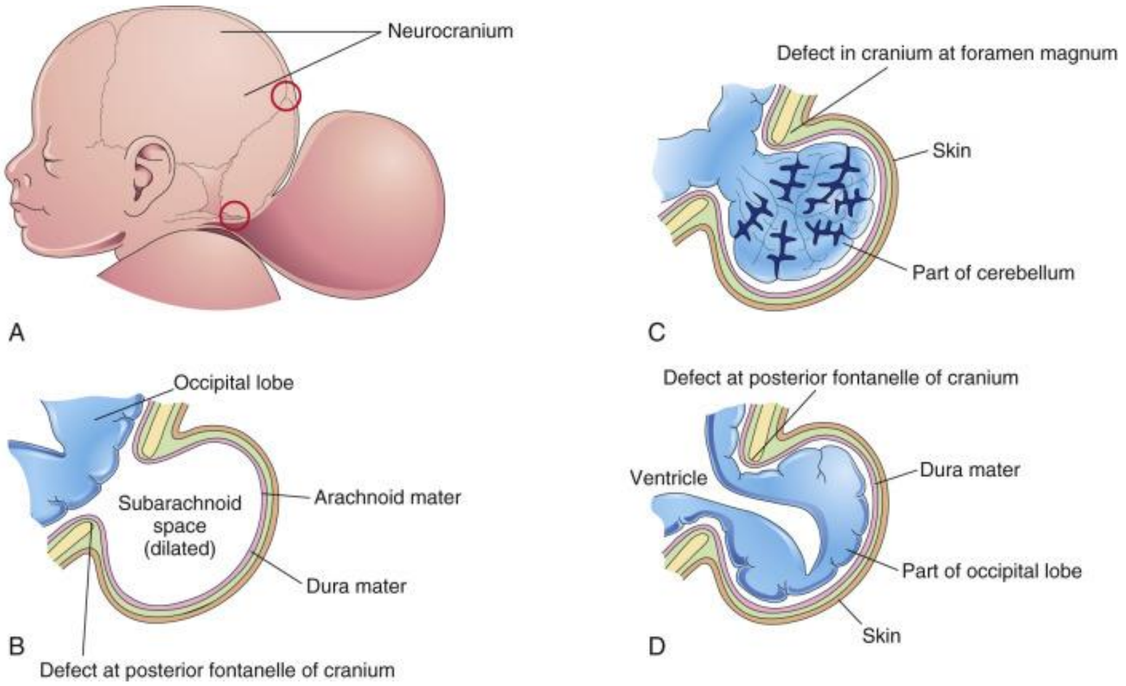
Failure of Caudal Neuropore Closure: Spina Bifida



Moore, The Developing Human, Fig. 17.12

A. Spina bifida occulta—often found incidentally with no symptoms
B. Meningocele
C. Meningomyelocele (myelomeningocele)
D. Myeloschisis

Failure of Rostral Neuropore Closure



Moore, The Developing Human, Fig. 17.31

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Encephalocele (cranium bifidum)—
a herniation of cranial contents (most common in occipital region)

Different degrees depending on herniation of neural tissue:

B: Meningocele

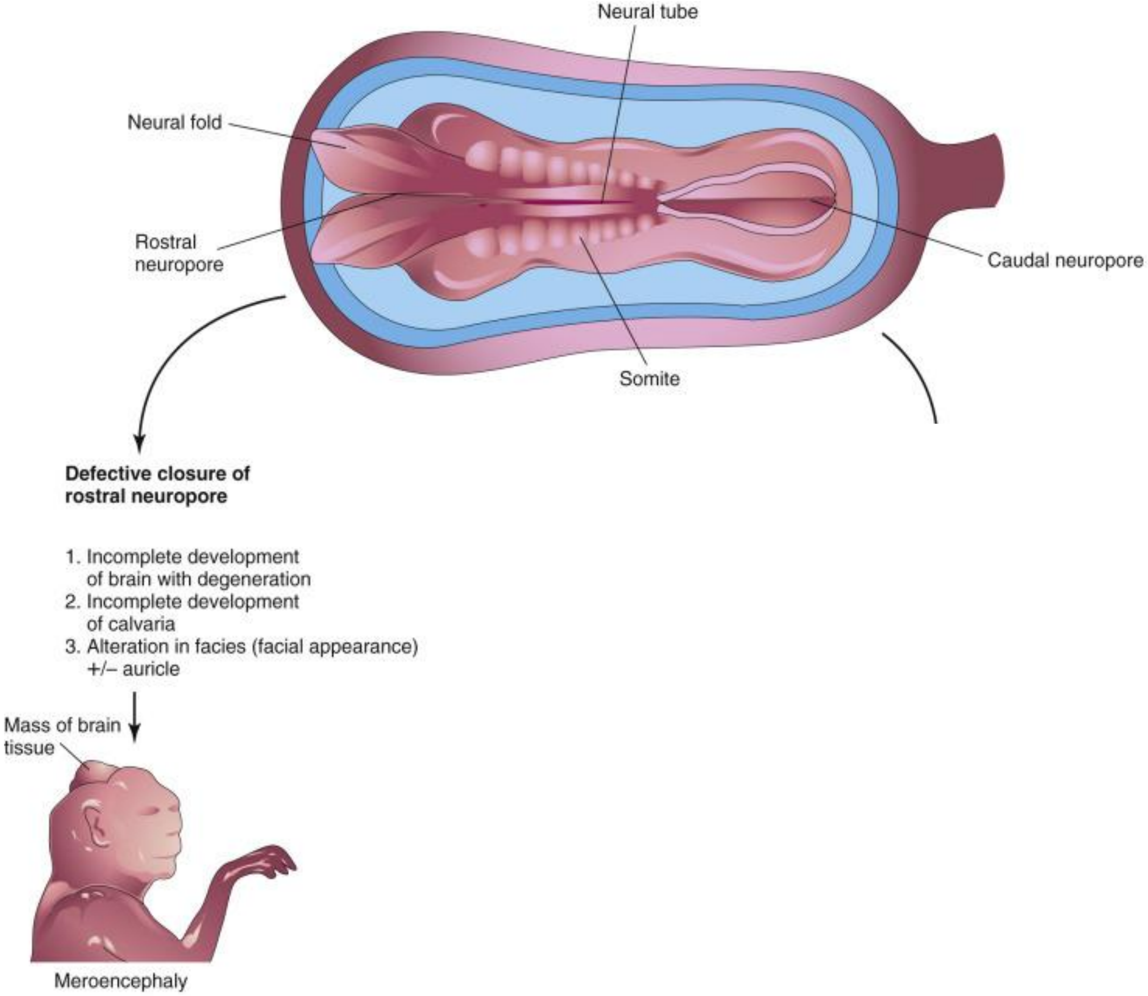
C: Meningoencephalocele

D: Meningohydroencephalocele

Failure of Rostral Neuropore Closure

The most severe cases result in anencephaly (also called meroencephaly). From this point, the forebrain and overlying skull fail to develop.

This is the equivalent of myeloschisis for spina bifida, with exposed neural tissue.

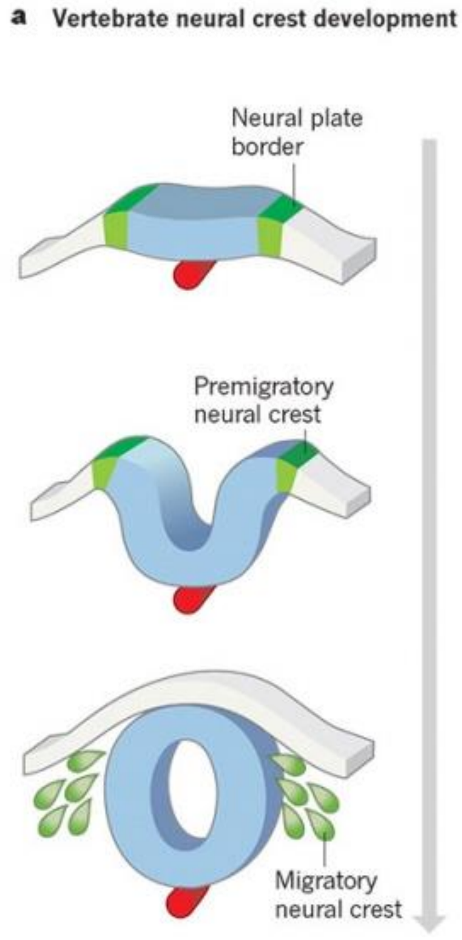


Moore, The Developing Human, Fig. 17.17

NEURAL CREST CELLS

Originate as ectoderm along the edge of the neural groove (that is—the neural crest)

As the neural tube separates from the external portion of ectoderm, neural crest cells separate from the ectoderm and migrate to various locations…



Green SA, Simoes-Costa M, Bronner ME. Nature. 2015 Apr 23;520(7548):474-82.

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NEURAL CREST CELLS

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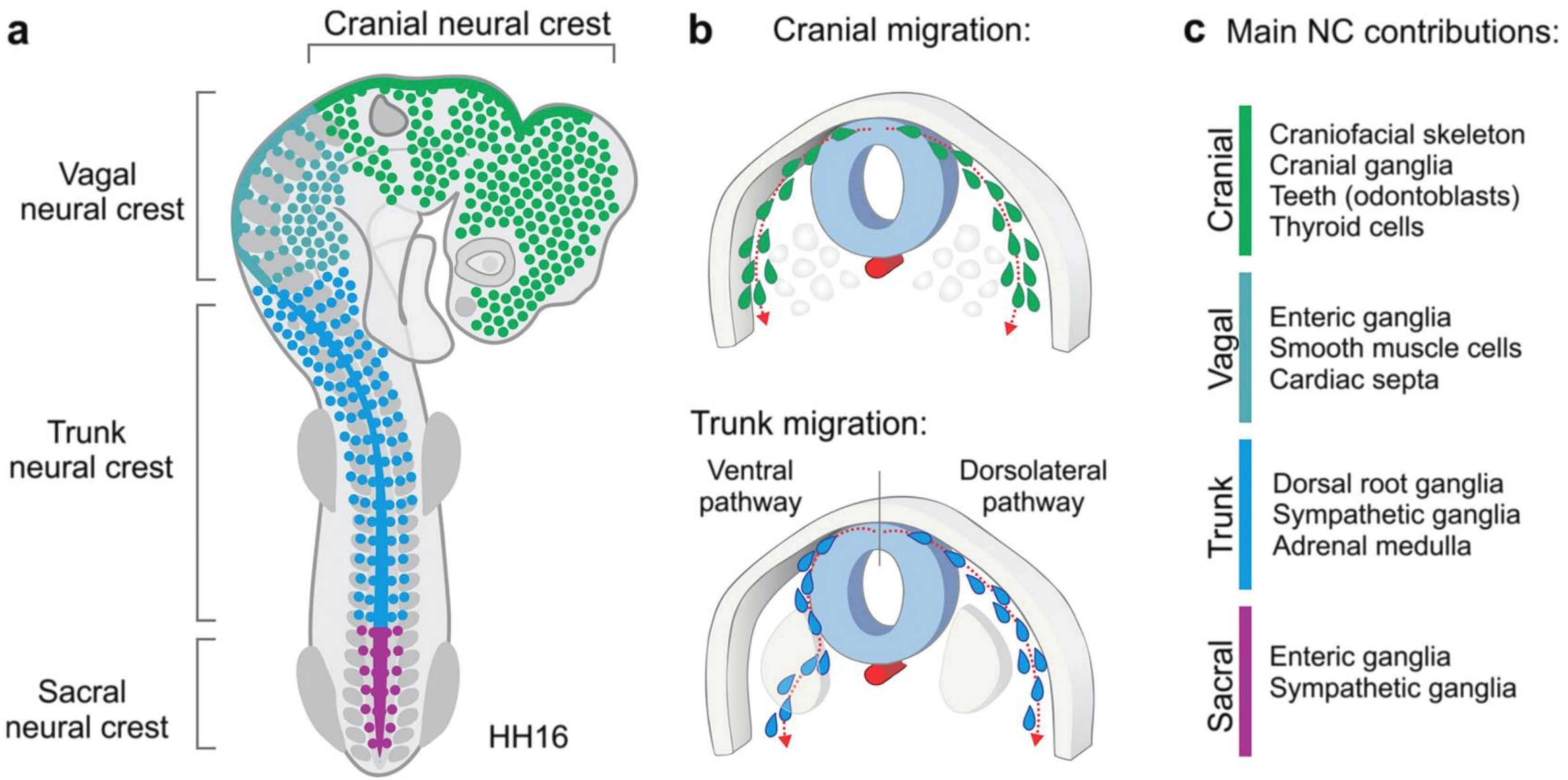
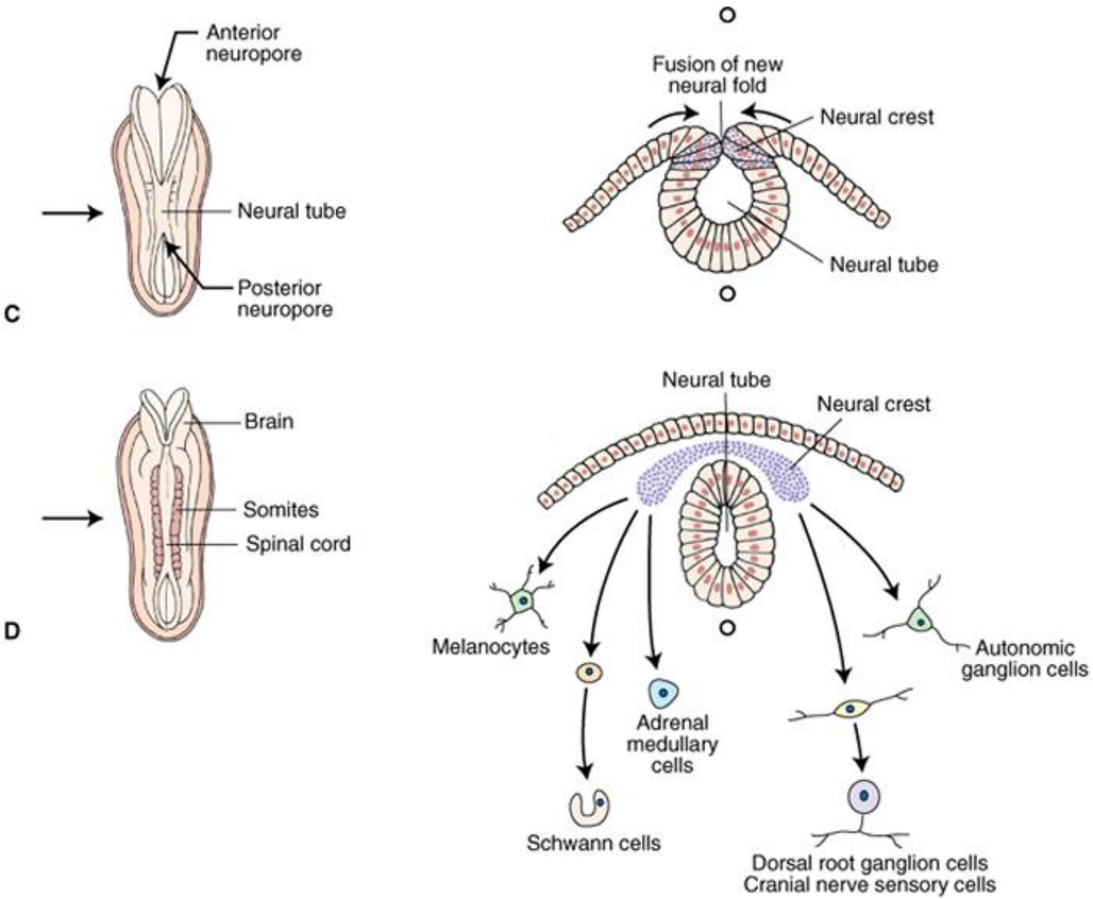


[Rothstein, et al. 2018. The molecular basis of neural crest axial identity.]()

Neural Crest Cells



Siegel and Sapru, Essential Neuroscience, Fig. 2.1

Migrate away from neural tube to form parts of peripheral nervous system:

- Dorsal root ganglia (i.e. somatic sensory neurons)
- Postganglionic autonomic neurons
- Some sensory ganglia of cranial nerves
- Arachnoid and pia mater
- Schwann cells

And also:

- Melanocytes
- Some facial bones
- Odontoblasts→teeth
- Laryngeal and tracheal cartilage
- Aorticopulmonary septum and parts of heart valves

Neural Tube Differentiation

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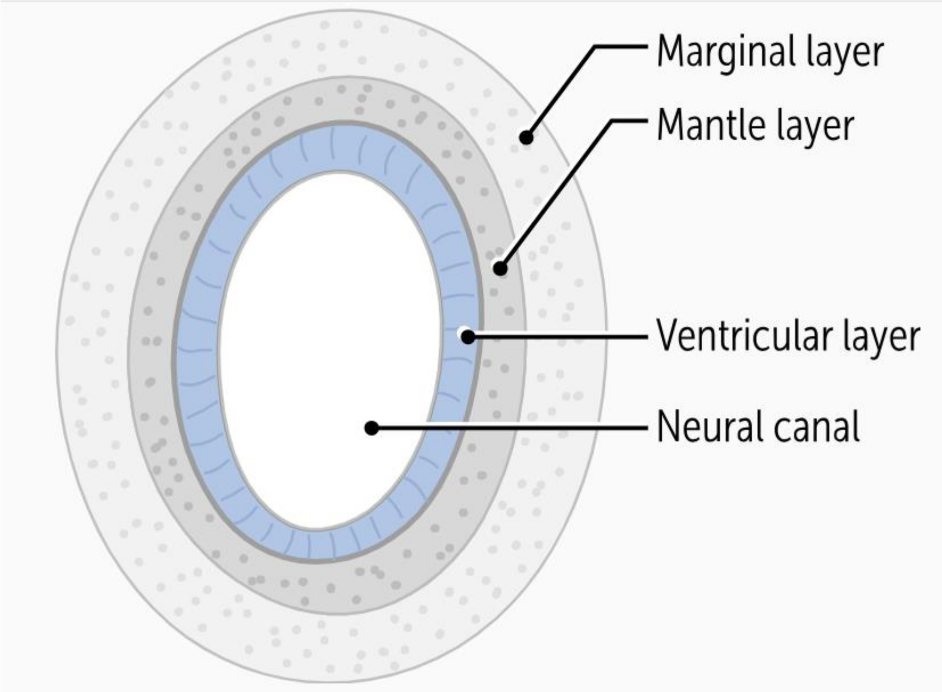

Marginal layer: becomes white matter

Mantle layer: becomes gray matter

Ventricular layer: becomes ependymal (glial) cells

Neural canal: becomes ventricles and central canal

Image from ScholarRx:
https://exchange.scholarrx.com/brick/development-of-the-nervous-system

Neural Tube Differentiation

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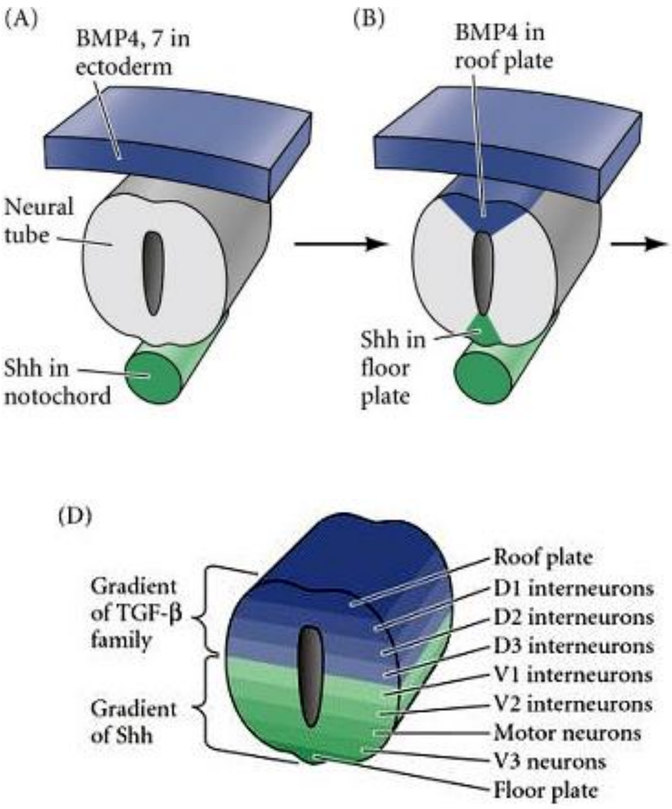
Concentration gradient formed by sonic hedgehog (**SHH**) and bone morphogenic proteins (**BMPs**) controls differentiation of neurons in the neural tube.

SHH secreted by the **notochord** induces neurons to develop near the floor plate (ventral)

BMP 4 and 7 signal the roof plate to secrete tissue growth factor (TGF-β).

Dorsal: TGF-β > SHH
sensory

Ventral: SHH > TGF-β
motor



Development of the Spinal Cord

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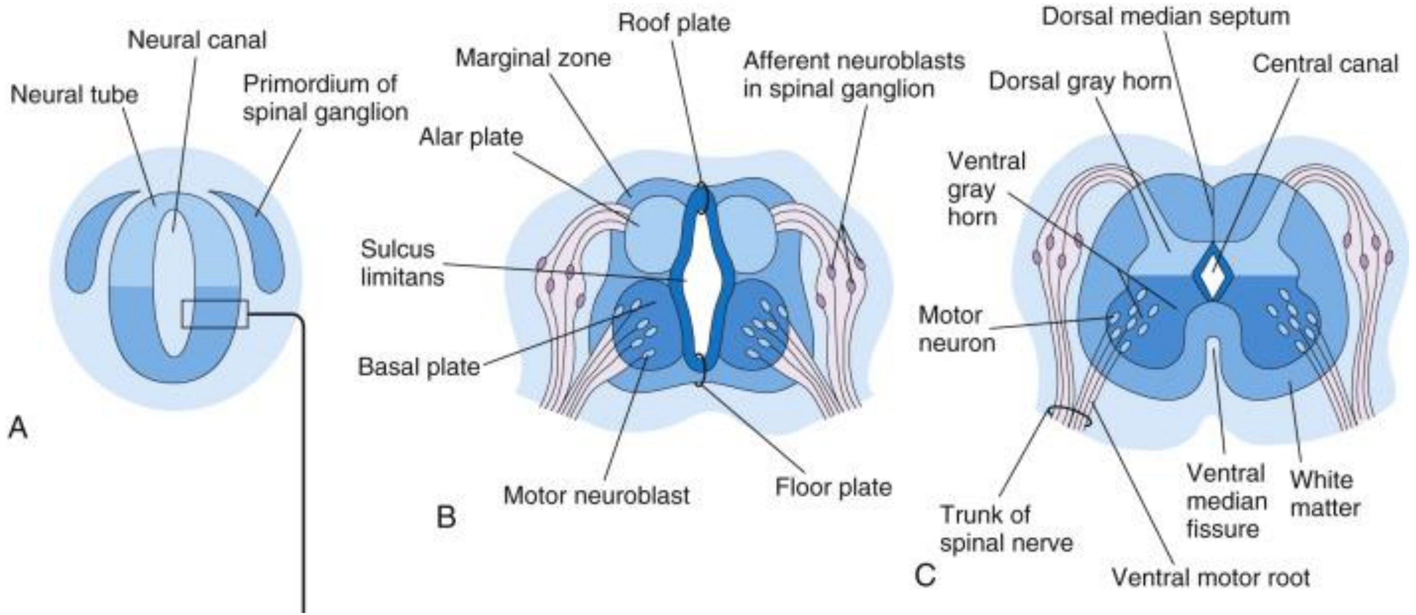

23 days

6 weeks

9 weeks

Moore, The Developing Human, Fig. 17.5

PERIPHERAL SENSORY VS. MOTOR NEURONS

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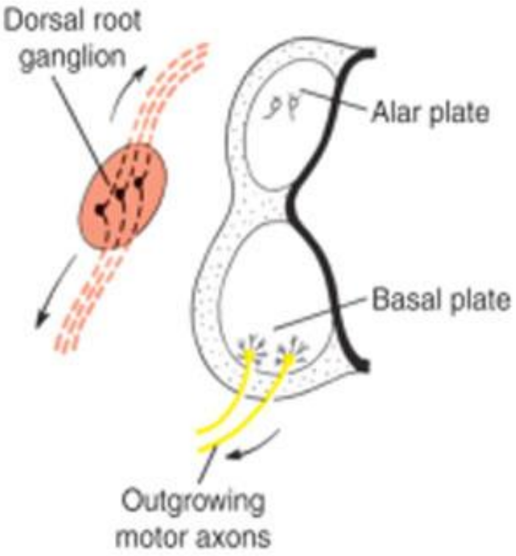
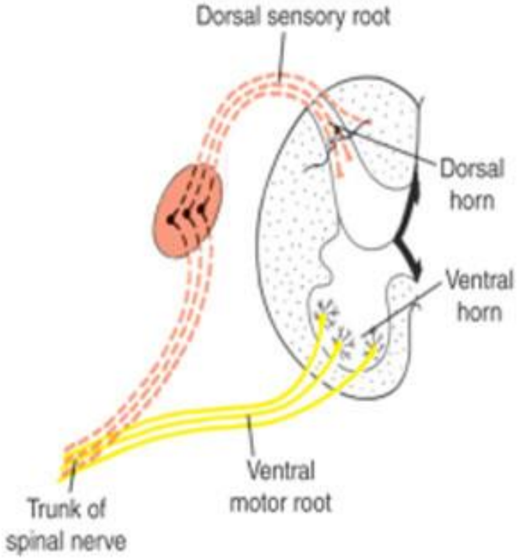




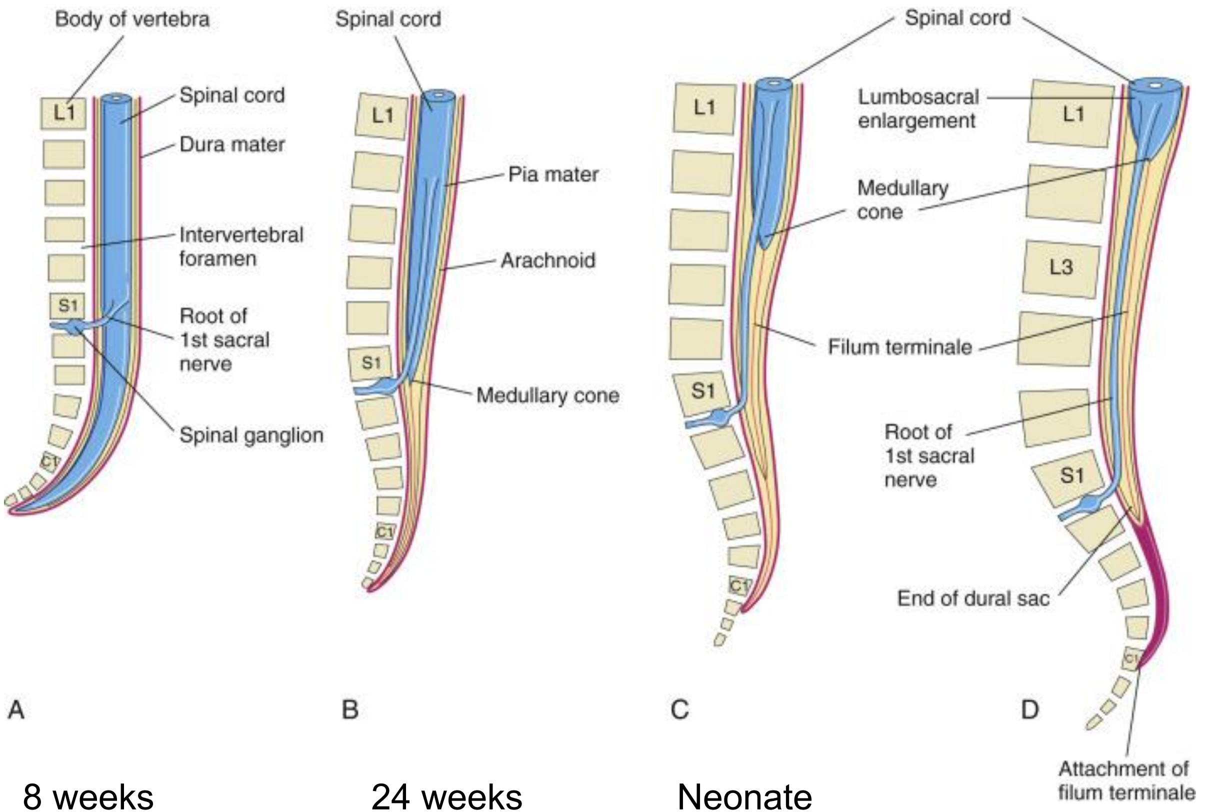
Sensory cell bodies are found in the dorsal root ganglia, OUTSIDE the central nervous system.

These cells develop from neural crest.

Motor cell bodies are found in the ventral horn of the spinal cord, INSIDE the central nervous system.

These cells develop from the neural tube.

Development of the Spinal Cord



8 weeks

24 weeks

Neonate

Adult

Moore, The Developing Human, Fig. 17.10

Vertebrae (and surrounding somites) grow faster than the spinal cord.

This shifts the location of the conus medullaris through development.

Note that with spina bifida, the spinal cord can be attached to surrounding tissue, which stretches the spinal cord during growth. This is **tethered cord syndrome**.

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Overview of Neuroembryology

Part 2: Brain

Karen Poole, PhD
Department of Biomedical and Anatomical Sciences
kpoole@nyit.edu

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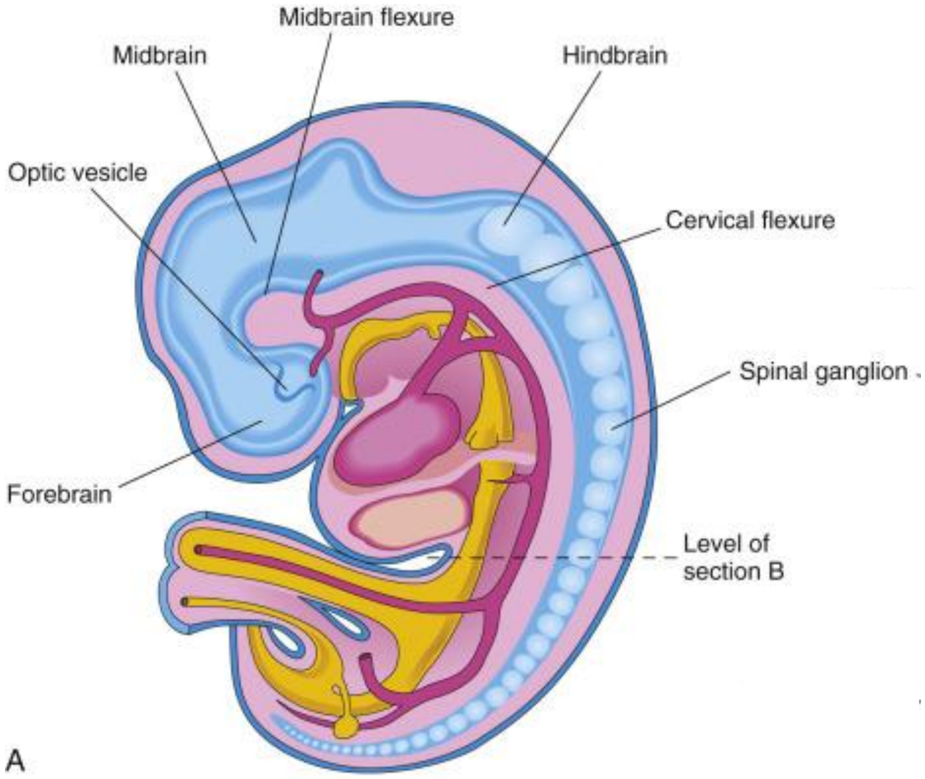
Development of the Brain



28 days

Moore, The Developing Human, Fig. 17.4

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Three primary vesicles:

Forebrain

Midbrain

Hindbrain

Development of the Brain

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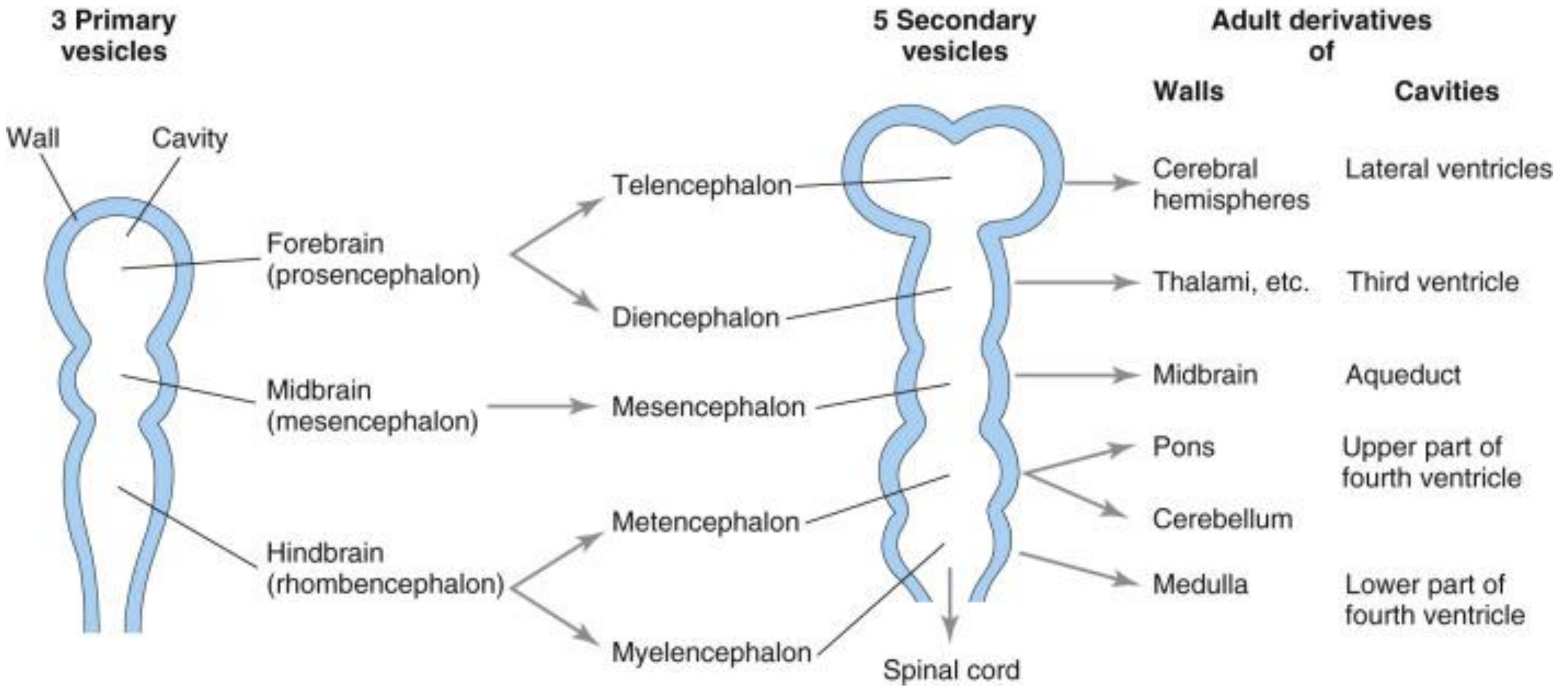

Moore, The Developing Human, Fig. 17.18

Development of the Brain

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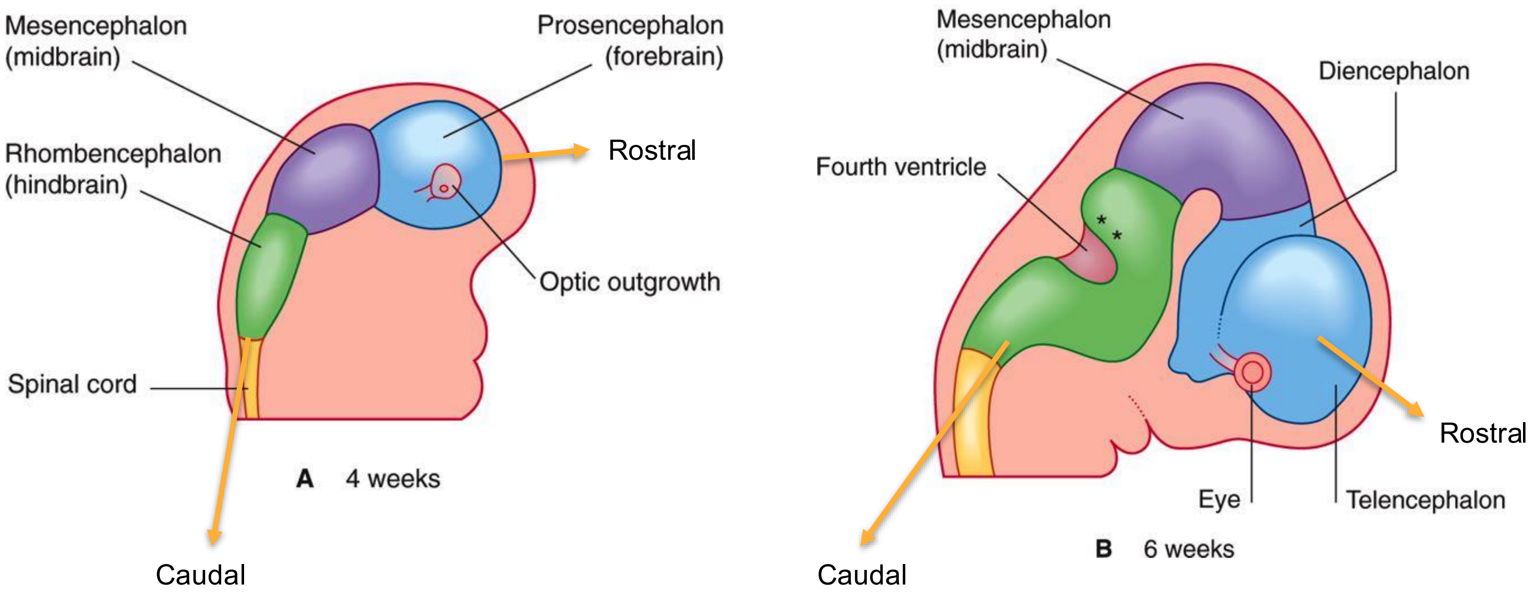
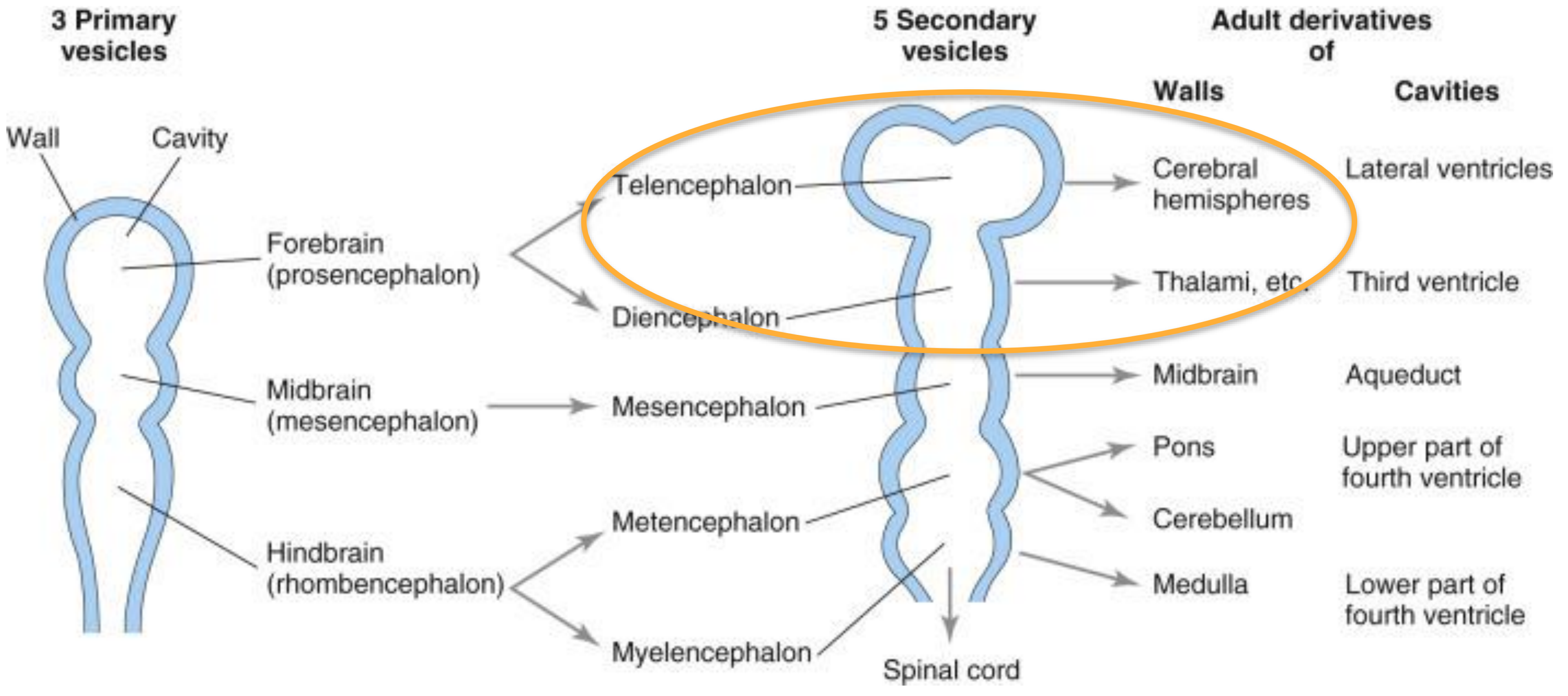


ADULT DERIVATIVES OF THE FOREBRAIN, MIDBRAIN, AND HINDBRAIN

Forebrain	Telencephalon	Cerebral hemispheres (neocortex) Olfactory cortex (paleocortex) Hippocampus (archicortex) Basal ganglia/corpus striatum Lateral and 3rd ventricles	Nerves: Olfactory (I)
	Diencephalon	Optic cup/nerves Thalamus Hypothalamus Mammillary bodies Part of 3rd ventricle	Optic (II)
Midbrain	Mesencephalon	Tectum (superior, inferior colliculi) Cerebral aqueduct Red nucleus Substantia nigra Crus cerebelli	Oculomotor (III) Trochlear (IV)
Hindbrain	Metencephalon	Pons Cerebellum	Trigeminal (V) Abducens(VI) Facial (VII) Acoustic (VIII) Glossopharyngeal (IX) Vagus (X) Hypoglossal (XII)
	Myelencephalon	Medulla oblongata	

Congenital Disorder: Holoprosencephaly



Moore, The Developing Human, Fig. 17.18

Congenital Disorder: Holoprosencephaly

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Complete or partial failure of the prosencephalon (forebrain) to divide along midline.

Leads to impaired lateralization of the forebrain (ventricles, cerebral hemispheres). Complete or partial absence of corpus callosum

Phenotypic: facial abnormalities, may see a single midline eye (cyclopia), tube-like nose with single nostril, cleft lip/palate, etc.

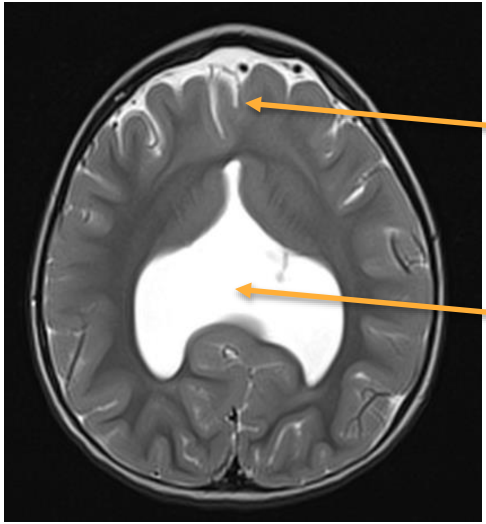
Severe cases may be stillborn and have high mortality in the first month of life.

Milder cases may experience seizures, motor impairment, hydrocephalus, hormone dysregulation

Causes variable, but include:

- Fetal alcohol syndrome
- Patau syndrome (trisomy 13)

T2 axial MRI of 2 year old girl



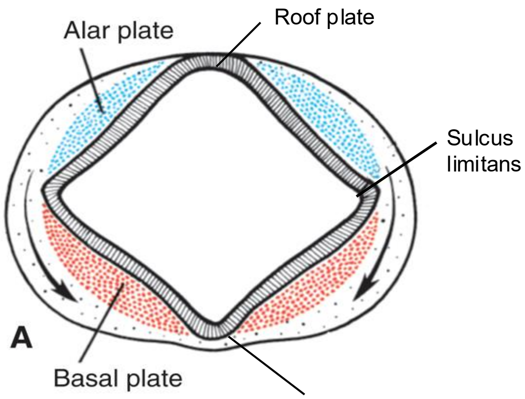
Longitudinal fissure not present anteriorly

Lateral ventricles joined in midline

Case courtesy of Ian Bickle, [Radiopaedia.org, rID: 23218]

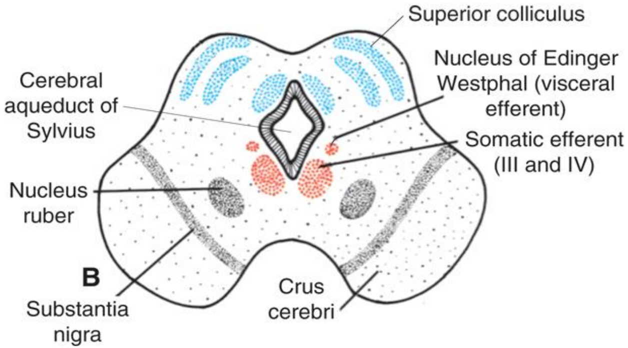
Brainstem vs. Spinal Cord

Early brainstem

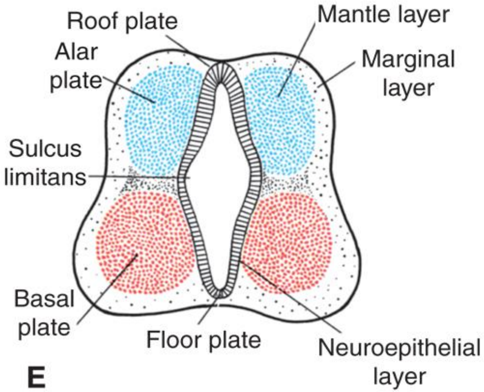


9-5A Mesencephalon development and aqueduct

Adult brainstem (midbrain)

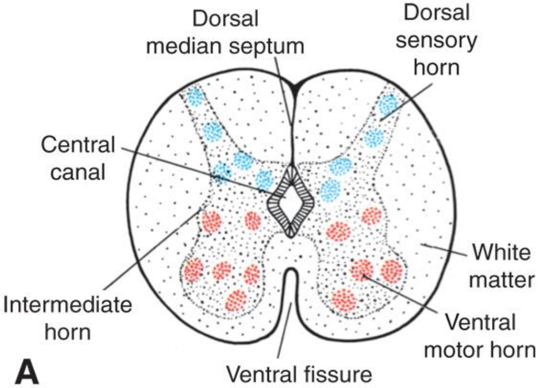


Early spinal cord



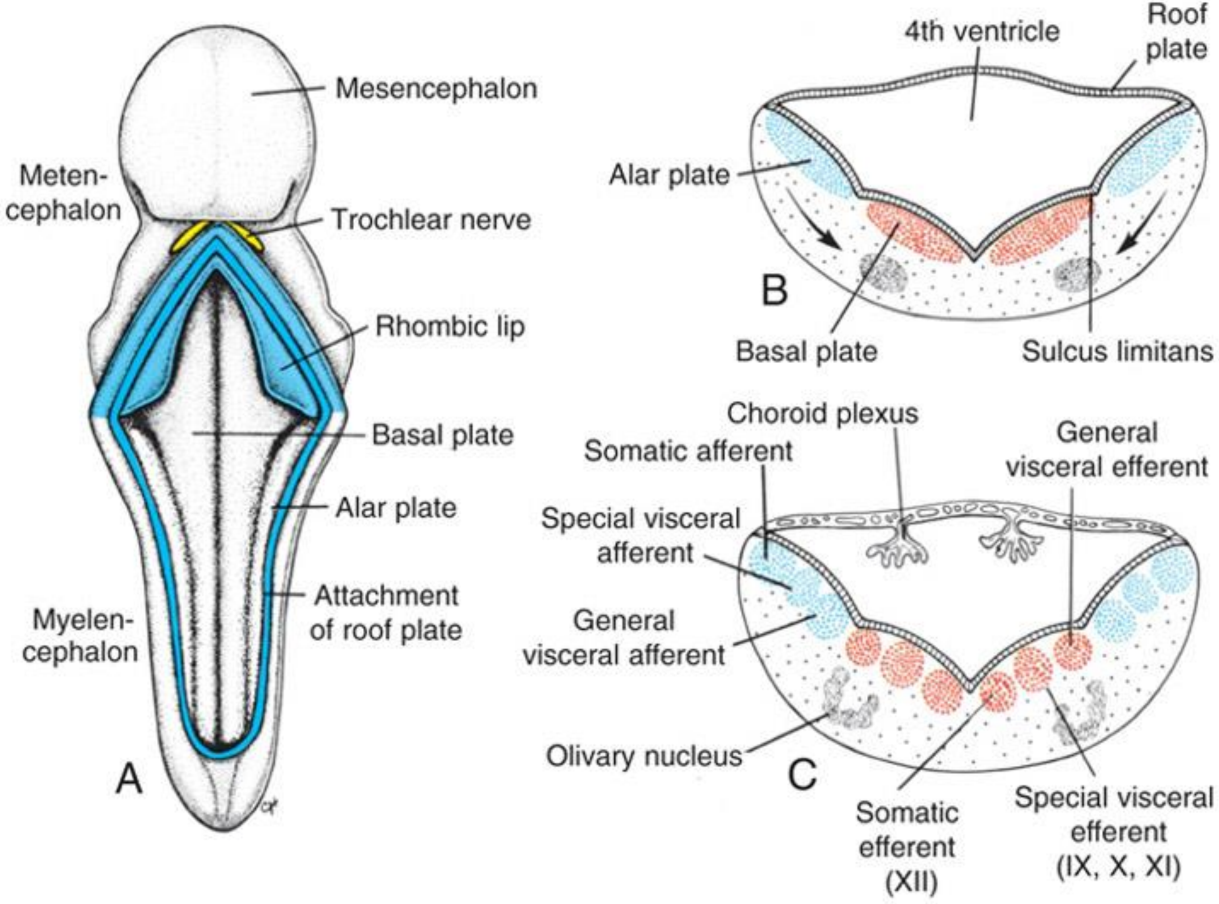
9-1E Spinal cord: Early cell differentiation and organizatior

Adult spinal cord



9-2A Nerve roots

MYELENCEPHALON

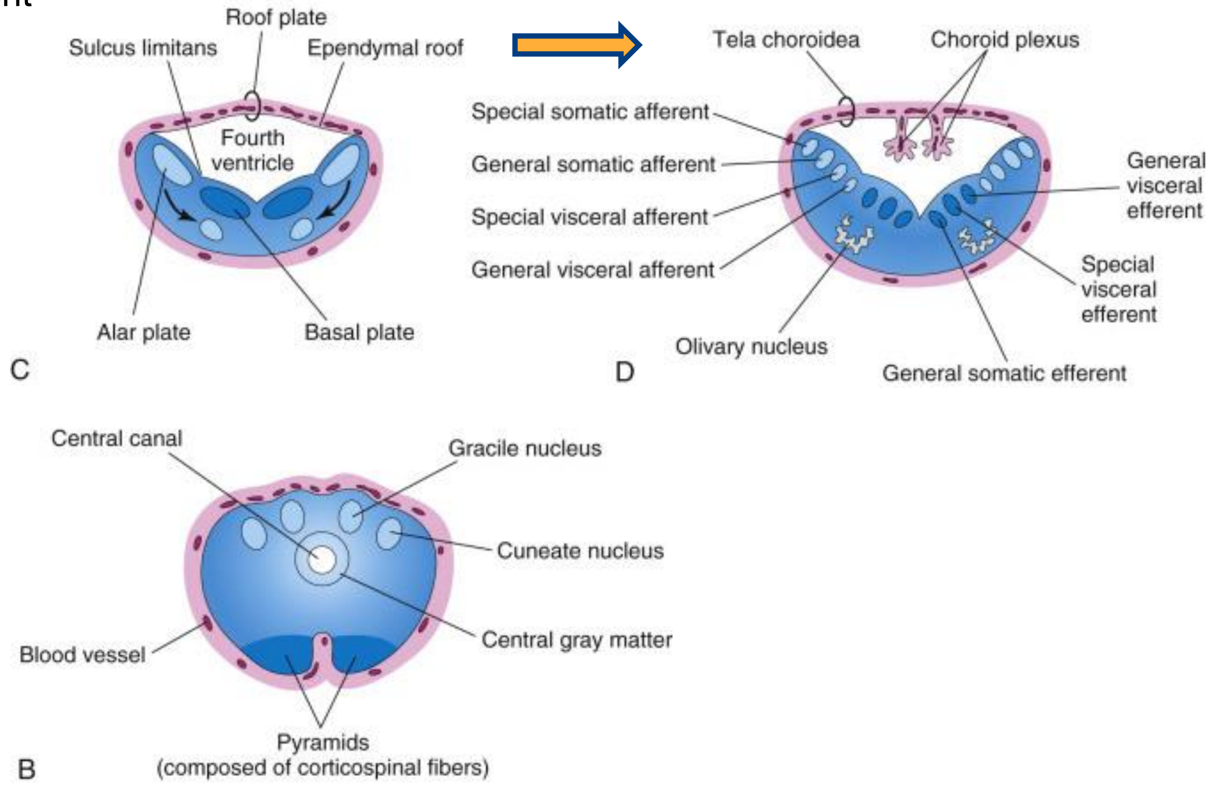


MYELENCEPHALON

5th week of development

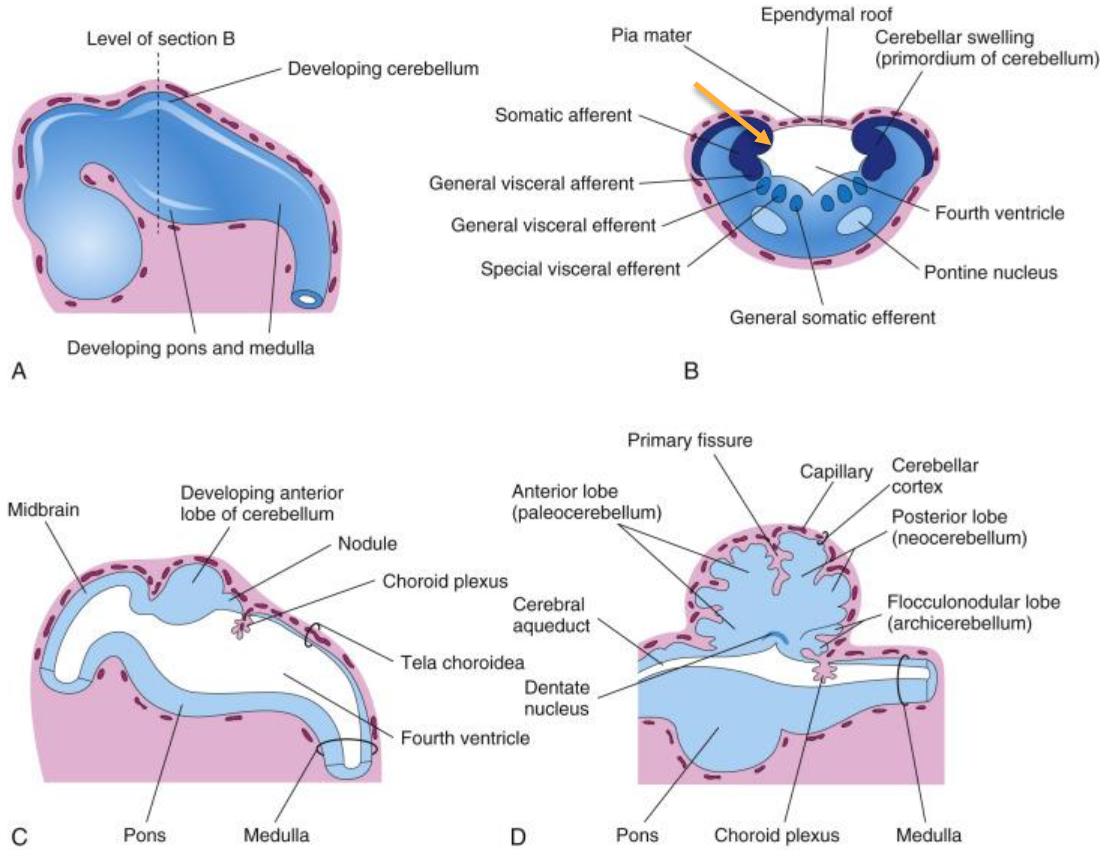
Upper (rostral) medulla

Lower (caudal) medulla



Moore, The Developing Human, Fig. 17.19

METENCEPHALON



Moore, The Developing Human, Fig. 17.20

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A, B, C: 5th week of development

D: later

Yellow arrow indicates rhombic lip

Congenital Disorder: Dandy-Walker Syndrome

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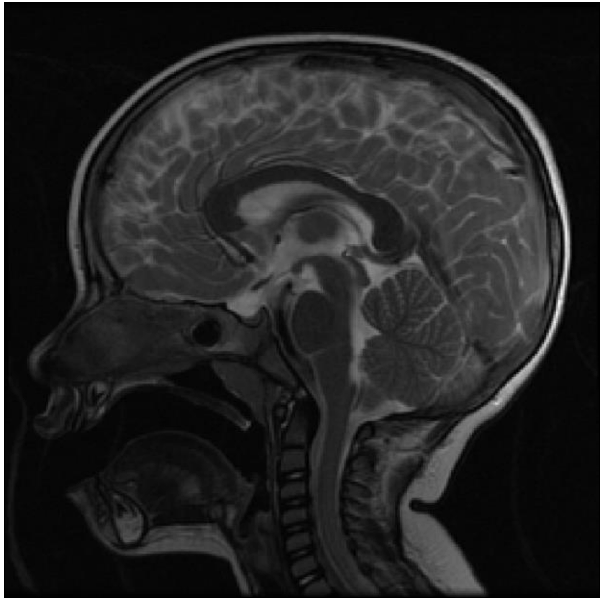
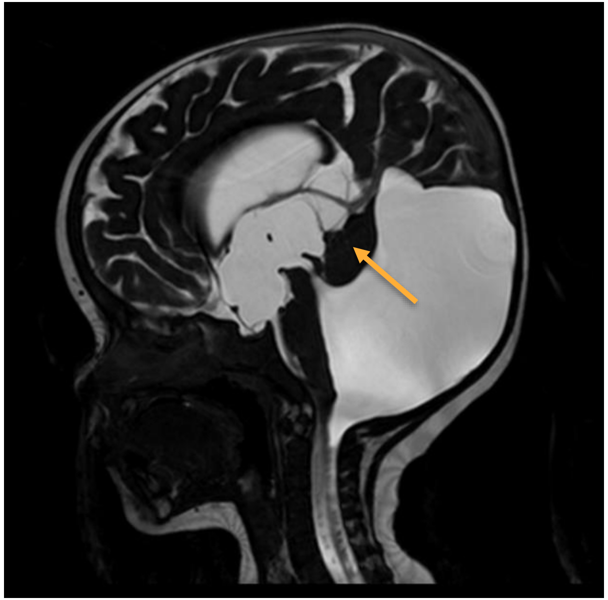
Normal midsagittal T2 MRI (patient is 2 years old)



Case courtesy of Bruno Di Muzio, Radiopaedia.org, rID: 43048

Midsagittal T2 MRI of 2 year old with Dandy-Walker Malformation



Case courtesy of Frank Gaillard, Radiopaedia.org, rID: 15808

Dandy-Walker Syndrome
Agenesis or hypoplasia of cerebellar vermis (yellow arrow) with cystic dilation of 4th ventricle

Ataxia—problems with coordination and/or unsteady gait

Often associated with non-communicating hydrocephalus (leading to increased intracranial pressure).
Signs: irritability, emesis

MESENCEPHALON

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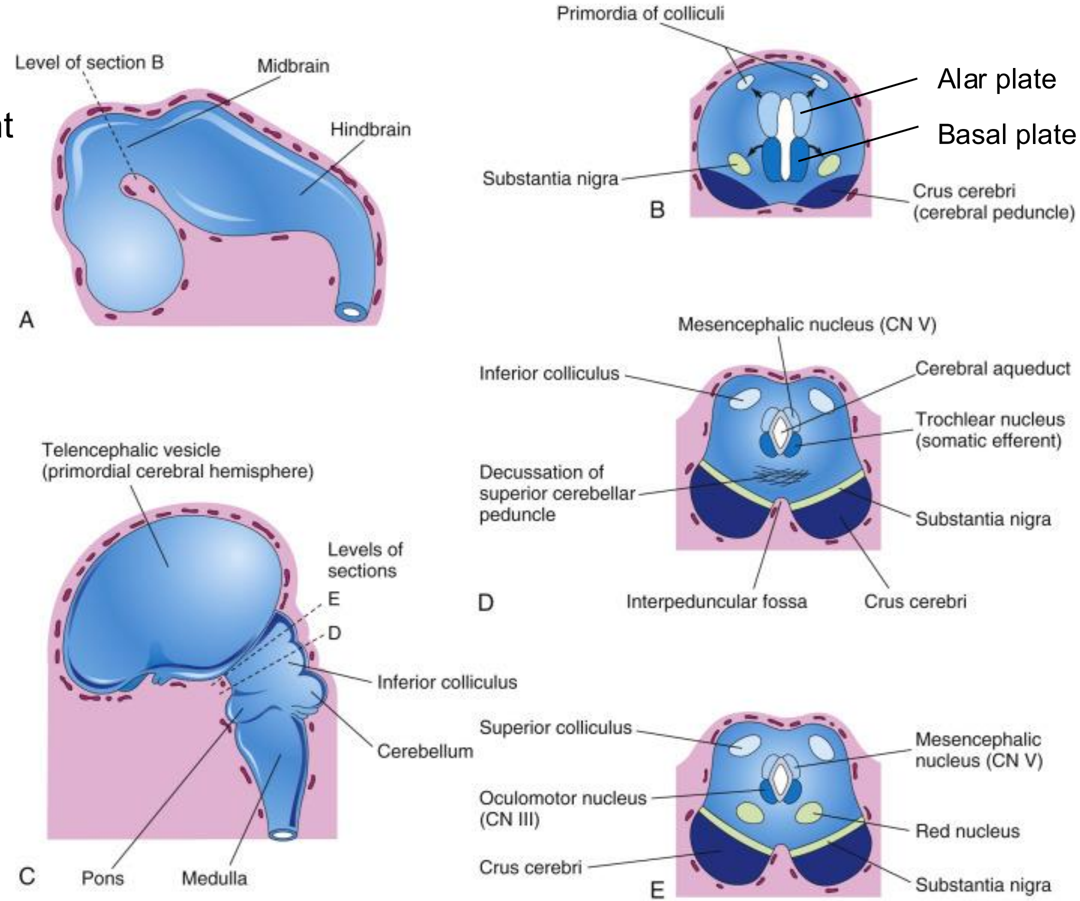
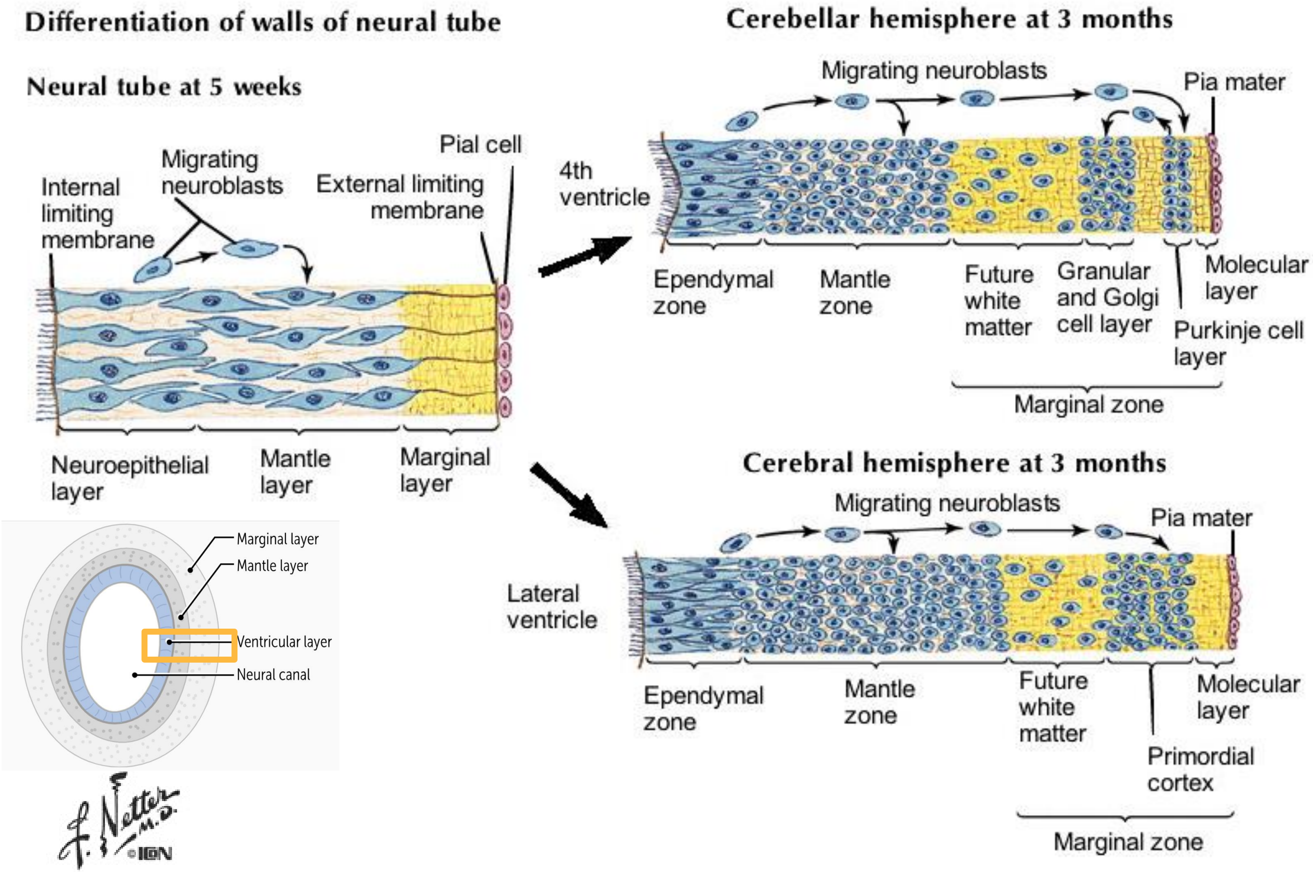
5th week of development

11th week of development



Moore, The Developing Human, Fig. 17.21

Migration of Neuroblasts



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In cerebellum and cerebrum, neuroblasts migrate outward, creating layers.

This is also why there is gray matter superficial/external to white matter tracts, unlike in the spinal cord.

Leads to growth of cerebrum and cerebellum.

Migration of Neuroblasts

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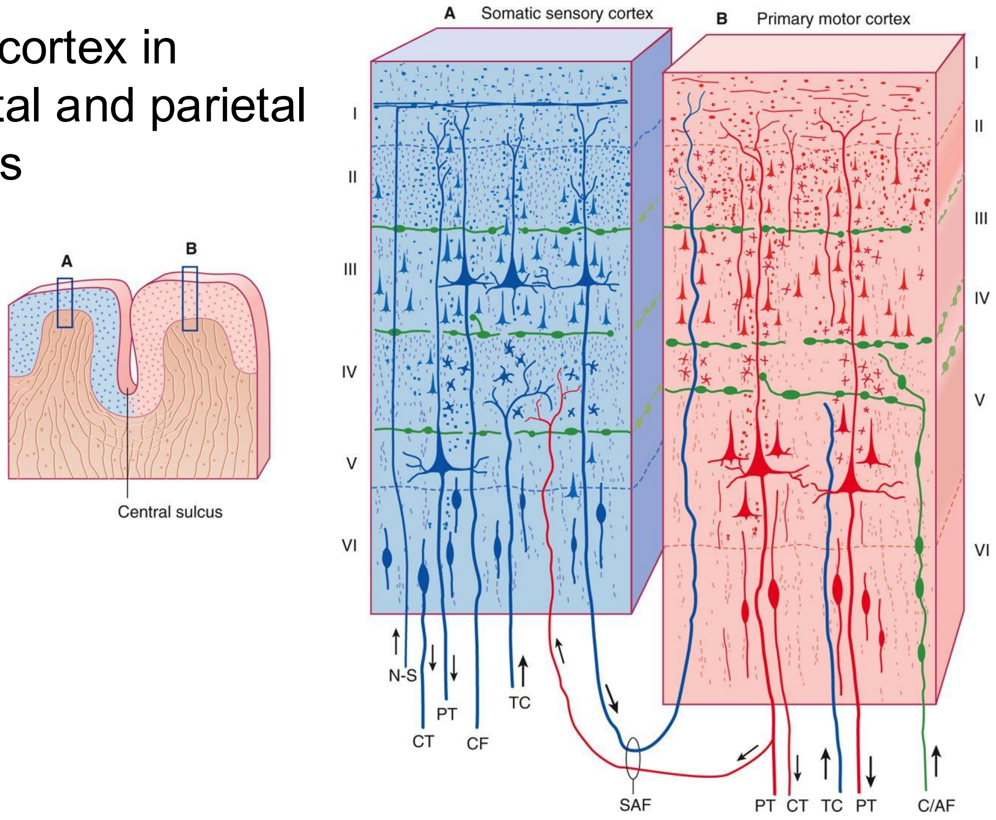
Neocortex in frontal and parietal lobes



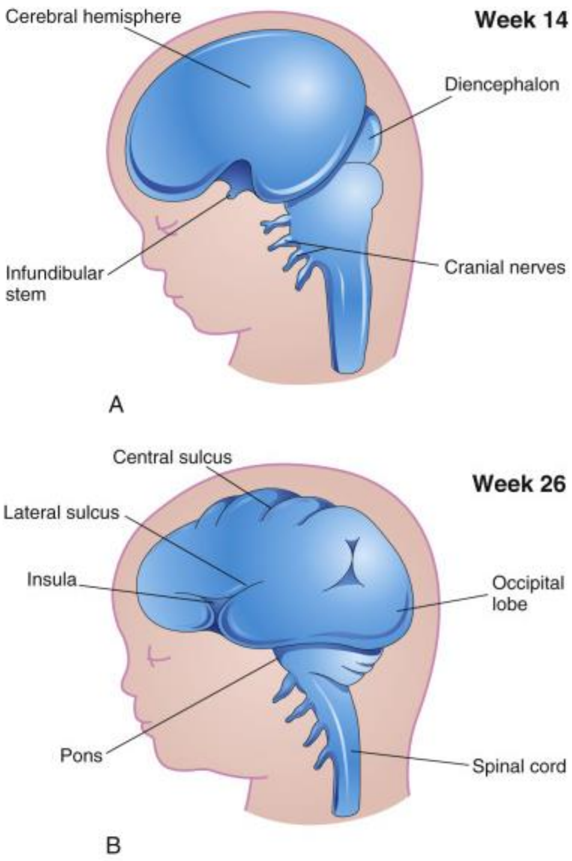
Clinical Neuroanatomy & Neuroscience, 6th Ed., Fig. 29.1

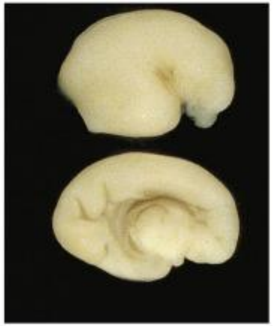
- Neocortex
  - Phylogenetically new
  - 6 layers
  - 90% of cortex
- Allocortex (Paleocortex)
  - Phylogenetically old
  - 3 layers
  - Found in olfactory regions and hippocampus

FETAL DEVELOPMENT

Through fetal development, the cerebrum grows and develops gyri and sulci as the number of neural cells increases.

This growth of the cerebrum also covers the diencephalon.





Moore, The Developing Human, Fig. 17.29

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Fetal Development

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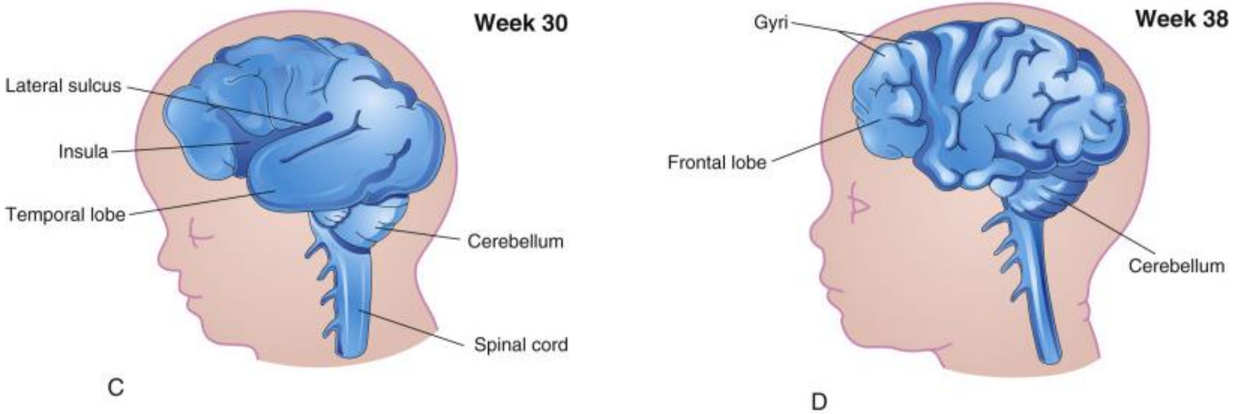
Note that the cortex of the insula becomes covered by the temporal lobe late in fetal develpment



Moore, The Developing Human, Fig. 17.29

CONGENITAL DISORDER: MICROCEPHALY

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Microcephaly—
reduction in neurogenesis. The lack of growth in the brain means the overlying bones of the calvaria also grow slowly

Multiple causes:
Genetic (e.g. autosomal recessive primary microcephaly)

Infectious agents (Zika, rubella, *Toxoplasma gondii*)

Exposure to alcohol (FAS)

Exposure to high amounts of ionizing radiation

Moore, The Developing Human, Fig. 17.36

Development of the Ventricles

Schematic view of ventricles

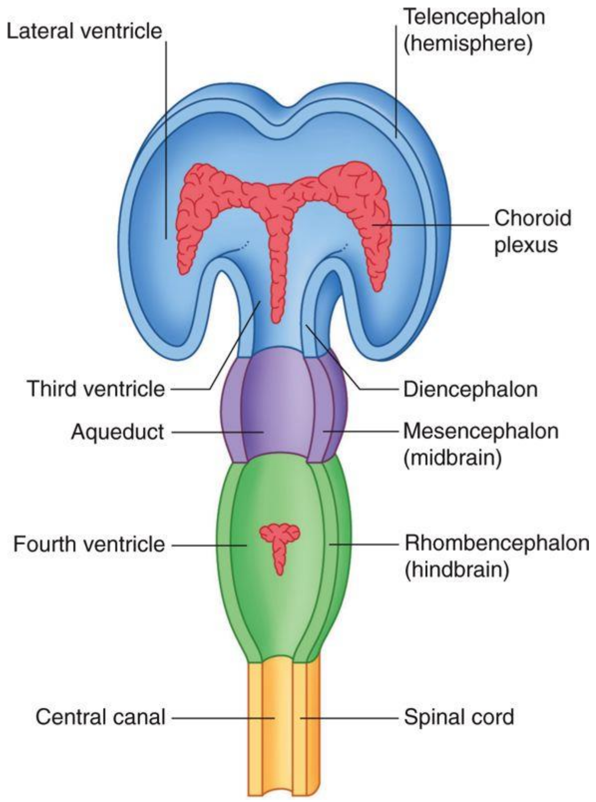

Fig. 1.5, *Clinical Neuroanatomy and Neuroscience, 6th Ed*

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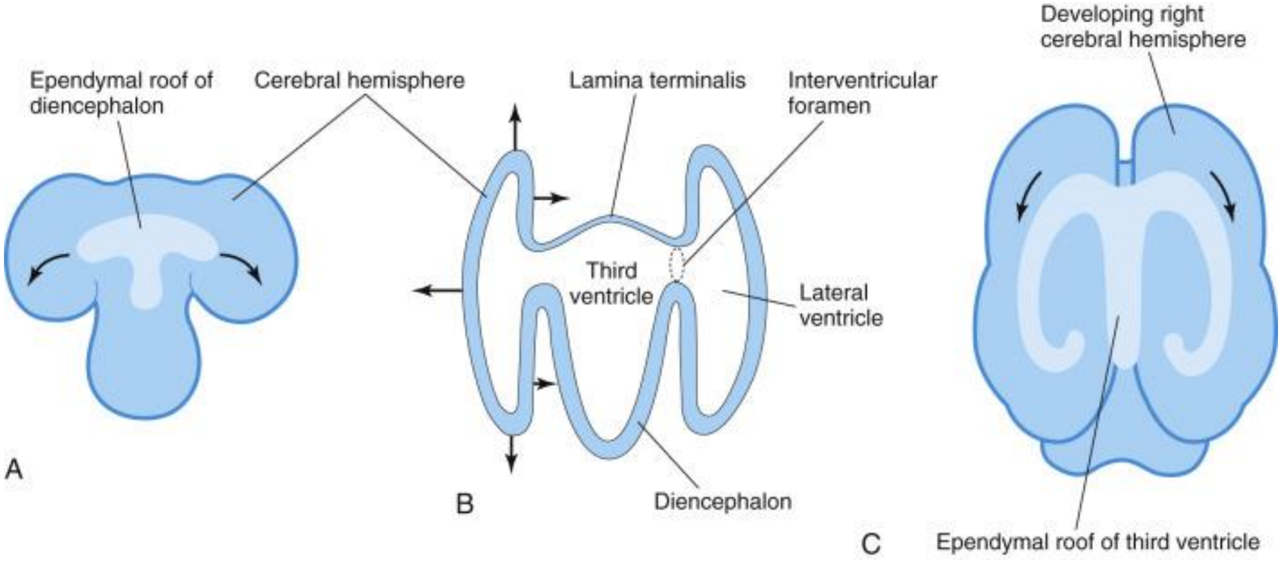
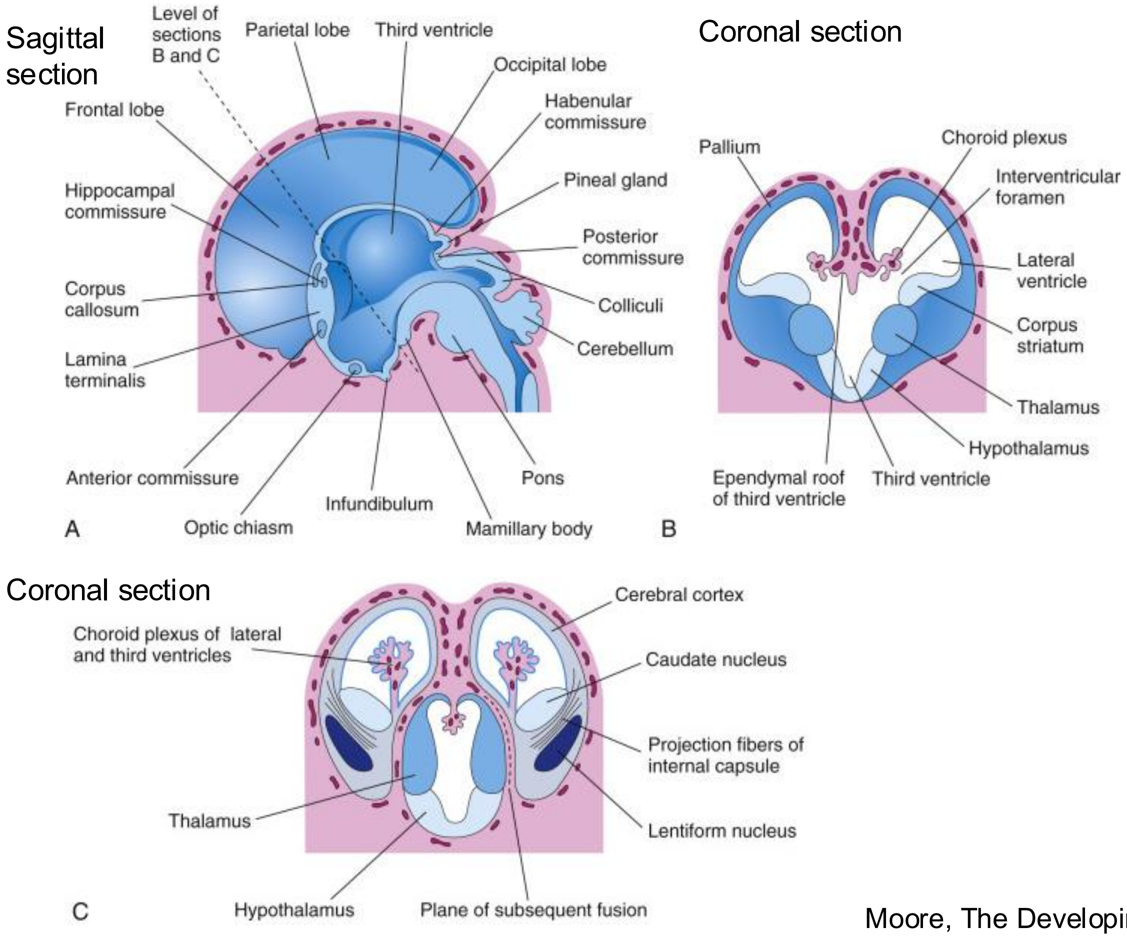
Development of the Ventricles



Moore, The Developing Human, Fig. 17.26

Development of the Ventricles



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Moore, The Developing Human, Fig. 17.27

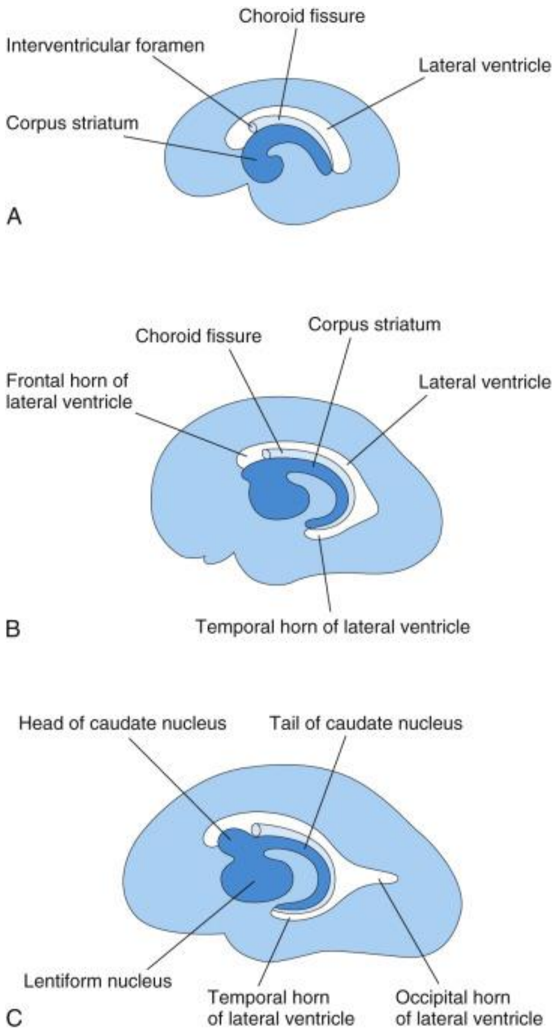
DEVELOPMENT OF THE LATERAL VENTRICLES

Lateral view

13 weeks

21 weeks

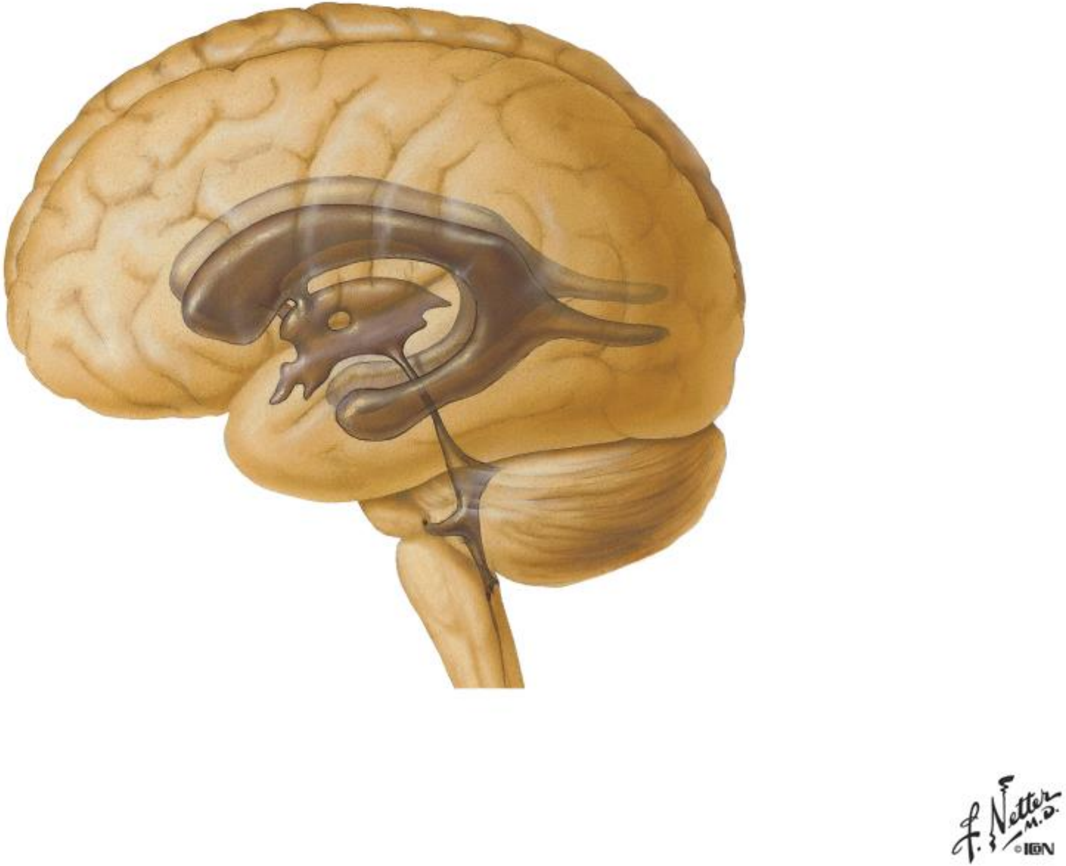
32 weeks



Light blue=cerebrum
Dark blue=basal ganglia/nuclei
White=lateral ventricle

Moore, The Developing Human, Fig. 17.28

FINAL CONFIGURAION OF THE VENTRICLES



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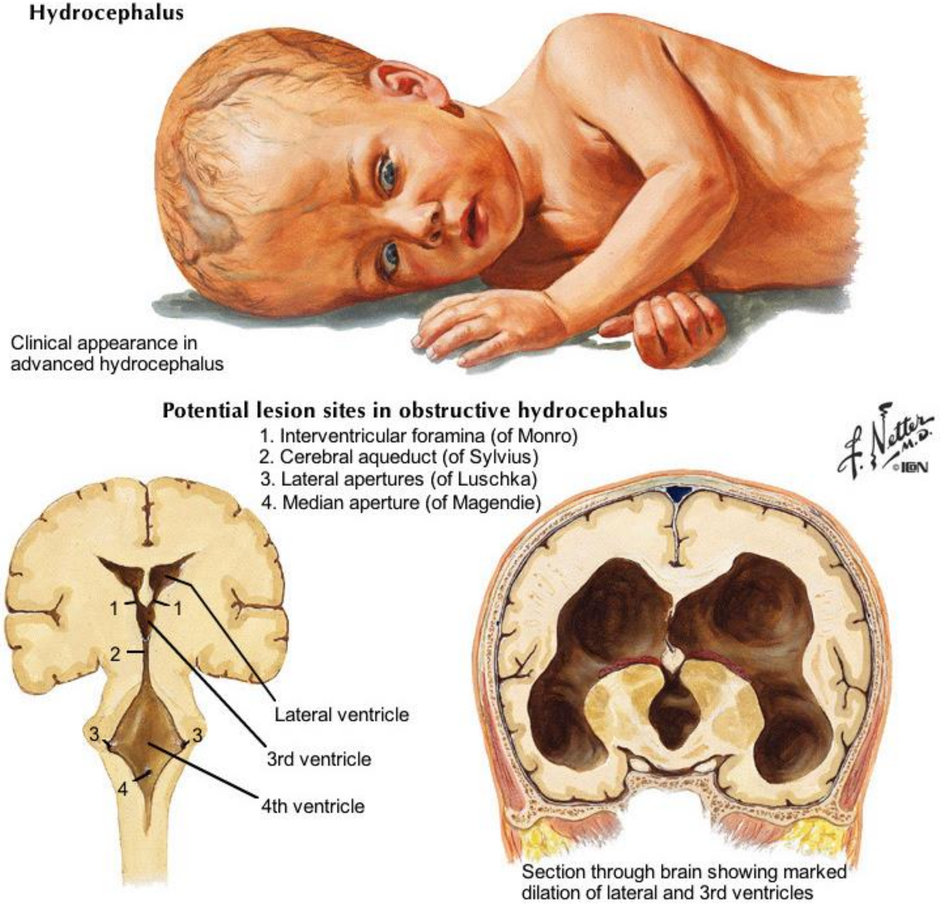
CONGENITAL DISORDER: HYDROCEPHALUS



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(Non-communicating) Hydrocephalus—
Can occur congenitally when ventricular system is not entirely open.

This happens most commonly in the cerebral aqueduct (congenital aqueductal stenosis).

Signs include large head size with bulging veins and fontanelles, irritability, drowsiness, vomiting, seizures.

Summary Slide

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- The nervous system develops from ectoderm
  - Neurulation is the process that forms the neural tube
  - Neural crest cells migrate from the neural tube to form parts of the peripheral nervous system and many other structures
- Basal plates form motor regions of the CNS and alar plates form sensory regions. Differences in development of the spinal cord and brain explain the anatomy of the CNS
- The brain is formed by areas of inflation of the neural tube called vesicles. Each vesicle gives rise to a portion of the brain and a portion of the ventricular system.

- Core References:
  - Siegel and Sapru, Essential Neuroscience, Chapter 2
  - Moore, Persaud, and Torchia, The Developing Human, 11th ed.
  - Scholar Rx: Development of the nervous system, Congenital disorders of the nervous system

Summary Slide

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Disorder	Embyronic structure affected	Result
Spina bifida	Caudal neuropore fails to close	Missing vertebral laminae, possible meningocele, meningomyelocele, or myeloschisis
Anencephaly	Rostral neuropore fails to close	Agenesis of much of brain and overlying bones
Encephalocele	Rostral neuropore fails to close/neural tube fails to separate from ectoderm	Opening in bones of calvaria, possible meningocele, meningoencephalocele, or meningohydroencephalocele
Holoprosencephaly	Prosencephalon fails to divide on midline into paired lobes of telencephalon	Lack of separate left and right cerebral hemispheres, agenesis/hypoplasia of corpus callosum
Dandy-Walker Syndrome	Rhombic lip fails to grow	Agenesis or hypoplasia of cerebellar vermis
Microcephaly	reduction in neurogenesis	Too few/simplified gyri of cerebrum
Hydrocephalus	Blockage/closure in ventricles	Enlarged ventricles, build up of CSF, large head size

Lecture Feedback Form:

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